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Sufficient Statistics for Nonlinear Tax Systems with General Across-Income Heterogeneity†

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This paper provides empirically implementable sufficient statistics formulas for optimal nonlinear tax systems in the presence of across-income heterogeneity in preferences, inheritances, income-shifting capabilities, and other sources. We characterize optimal smooth tax systems on income and savings (or other commodities), as well as simpler tax systems. We use familiar elasticity concepts and a novel sufficient statistic for heterogeneity correlated with earnings ability: the difference between across-income variation in savings and the causal effect of income on savings. We apply these formulas to the United States and find that the optimal savings tax is mostly positive and progressive. (*JEL* E21, G51, H21, H24)

Taxes on capital income, estates, inheritances, and certain categories of consumption are a widespread feature of modern tax systems. Yet there is considerable debate, both among economists and in policy circles, about their optimal design. The celebrated theorem of Atkinson and Stiglitz (1976) is sometimes interpreted to suggest that such taxes should be eliminated. The theorem states that if preferences are homogeneous and weakly separable in consumption and labor, then differential taxes on commodities—including on future consumption in the form of savings—are suboptimal, and welfare is maximized when redistribution is carried out solely through an income tax. However, as was appreciated by contemporaneous work (Mirrlees 1976) and emphasized by the authors themselves (Stiglitz 2018), the assumptions underpinning the Atkinson-Stiglitz Theorem are strong, limiting its relevance for the design of tax policy.

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It has been shown that differential commodity taxes may be optimal when there is heterogeneity in consumption preferences, inheritances, rates of return, or in the presence of complementarities between labor and certain forms of consumption.[1] However, this collection of seemingly disparate—and typically qualitative—results does not provide guidance for translating estimable empirical statistics to practical, quantitative policy implications. This is in contrast to the optimal income taxation literature, where optimal income tax formulas utilizing observable "sufficient statistics"—such as elasticities and income effects—have been influential in public finance (e.g., Diamond 1998; Saez 2001).[2]

In this paper, we derive sufficient statistics formulas for optimal linear and nonlinear commodity taxes in a setting with general across-income heterogeneity—e.g., in preferences, inheritances, or rates of return, all of which may vary with earnings ability—and where weak separability of the utility function may not hold. These formulas utilize familiar elasticity concepts and a novel sufficient statistic that captures many different sources of across-income heterogeneity. The formulas connect to modern empirical work measuring parameters such as the elasticity of capital income, marginal propensities to consume, heterogeneity in rates of return, and the distribution of wealth and savings (e.g., Fagereng et al. 2020; Fagereng, Holm, and Natvik 2021; Agersnap and Zidar 2021; Jakobsen et al. 2020; Smith, Zidar, and Zwick 2021; Golosov et al. 2021). We derive these results in a general version of standard models where consumers with heterogeneous earning abilities and preferences choose labor supply and a consumption and savings bundle that exhausts their after-tax income.[3] Our formulas nest prior results in this setting, as well as the Atkinson-Stiglitz Theorem itself, as special cases.

We organize the paper around the following key contributions. For concreteness, we describe results in terms of taxes on savings, although they also apply to other commodities.

First, we characterize optimal nonlinear taxes on earnings and savings that satisfy only smoothness restrictions, in settings with unidimensional and multidimensional heterogeneity. In the often-studied special case where heterogeneity is unidimensional, meaning that at each earnings level z there is a single savings level s, optimal tax formulas take a particularly simple form. Across-income variation of s with z, given by the total derivative $s'(z)$, can then be expressed as the sum of two partial derivatives: (i) the causal income effect $s'_{inc}(z)$, holding taxpayer type constant, and (ii) across-income heterogeneity in the degree to which higher-ability types prefer, or are able to obtain, more s, holding earnings constant. This second component, $s'(z) - s'_{inc}(z)$, is a key statistic summarizing preference (or other) heterogeneity that is correlated with earnings ability. This statistic can be estimated from

[1] Heterogeneity in consumption preferences: Saez (2002); Diamond and Spinnewijn (2011); Golosov et al. (2013); Gauthier and Henriet (2018). Inheritances: Boadway, Marchand, and Pestieau (2000); Cremer, Pestieau, and Rochet (2003); Piketty and Saez (2013). Rates of return: Gahvari and Micheletto (2016); Gerritsen et al. (2020); Schulz (2021). Complementarities between labor and consumption: Corlett and Hague (1953); Jacobs and Boadway (2014).

[2] See also Rothschild and Scheuer (2013); Stantcheva (2017); Sachs, Tsyvinski, and Werquin (2020); Hendren (2020); Bierbrauer, Boyer, and Hansen (2023).

[3] See, e.g., Atkinson and Stiglitz (1976); Saez (2002); Farhi and Werning (2010); Diamond and Spinnewijn (2011); Golosov et al. (2013); Piketty and Saez (2013); Scheuer and Wolitzky (2016); Saez and Stantcheva (2018); Allcott, Lockwood, and Taubinsky (2019); Gaubert, Kline, and Yagan (2021); Hellwig and Werquin (2022).

existing data on the correlational and causal associations with earnings, without the need to directly measure or model the relationship between earnings ability and difficult-to-observe attributes like preferences. We show that the formula for optimal savings tax rates is a product of $s'(z) - s'_{inc}(z)$ and a term that resembles the optimal income tax formula in Saez (2001), with the elasticity of earnings replaced by the elasticity of savings. This result generalizes both the Atkinson-Stiglitz Theorem and qualitative results about when that theorem does not hold: the optimal savings tax rate is everywhere zero if and only if $s'(z) = s'_{inc}(z)$ for all earnings z. Consequently, this formula can be viewed both as a synthesis of prior work and a practical, empirically oriented guide for optimal tax design. When heterogeneity is multidimensional, meaning that savings are heterogeneous at each earnings level, the formula for optimal savings tax rates generalizes the unidimensional result. The formula then depends on how the *distribution* of savings varies with earnings and how much of that across-income variation can be explained by causal income effects, s'_{inc}. Additionally, novel income-conditional covariance terms—such as the income-conditional covariance between social marginal welfare weights and savings—clarify the way in which multidimensional heterogeneity alters optimal tax rates.

Prior work on mechanism design with across-income heterogeneity has emphasized a tagging logic for taxing s. For instance, if higher-ability types prefer more s, then consumption of s reveals information about earnings ability that can be exploited with taxes on s. However, this insight is difficult to implement because it is challenging to measure preferences and ability directly. Our approach provides an empirical strategy for quantifying the optimal tax on s that does not require direct measurement of the primitives. We show that the degree by which a tax on s distorts earnings is proportional to the causal income effects s'_{inc}, implying that the tax is more attractive if s'_{inc} is small relative to the across-income variation in s. Importantly, this result holds for a variety of channels that might drive across-income variation in s, including heterogeneous endowments, differential rates of return on investments, deviations from weak separability, differential capacities to engage in income shifting, and some models of bequests and human capital investments.

Our second contribution is a characterization of what we call "simple tax systems." In many countries, the tax system consists of a nonlinear tax on earnings, accompanied by taxes on savings vehicles that can be classified as one of three types: (i) a separable linear savings tax; (ii) a separable nonlinear savings tax; or (iii) a system with a linear earnings-dependent savings tax, which allows, for example, lower-income people to have their savings taxed at a lower linear rate, as is the case for long-term capital gains in the United States. We show that the optimal tax policy within each of these classes of simple systems can be expressed using the same sufficient statistics that appear in our formulas for unrestricted smooth tax systems. We also characterize optimal savings taxes when the earnings tax is fixed and not necessarily optimal.

We then extend our results to several other applications. First, we consider situations where the government wants to correct individual savings behavior because of externalities or behavioral biases. Our model generalizes the setup of Farhi and Werning (2010), in which the government puts more weight on future generations than the parents, to allow for heterogeneous preferences. Our results also cover the case where individuals undersave due to behavioral biases such as myopia or lack of self control, as in Moser and Olea de Souza e Silva (2019), or where capital accumulation through savings triggers macro externalities, as in Howitt and Aghion (1998).

Our second extension studies settings in which there is an additional efficiency rationale for taxing savings because the government can collect savings taxes either before or after returns are earned and can therefore arbitrage heterogeneity in rates of return by shifting tax collections to post-returns savings for individuals with high returns. This extension relates to independent work by Gerritsen et al. (2020), who study the special case where all across-income heterogeneity is from differences in rates of return, characterizing the optimal separable nonlinear savings tax in terms of model primitives. Additionally, we consider in the online Appendix a more general setup with many dimensions of consumption and savings. This allows us to consider applications where, for example, different categories of savings might be taxed differently.

In the final part of the paper, we use our sufficient statistics formulas to study the optimal tax treatment of savings in the United States in a two-period model where savings accumulated during the working life are consumed in retirement (Saez 2002; Golosov et al. 2013). To get at $s'(z)$, we calibrate the distribution of savings across incomes using the Distributional National Accounts micro-files of Piketty, Saez, and Zucman (2018). The data suggest that savings are approximately constant at low incomes but increase convexly at higher incomes. To calibrate the causal income effects on savings s'_{inc} at the individual level, we draw on three sources. The first is Fagereng, Holm, and Natvik (2021), which estimates the medium-run marginal propensity to save out of windfall income using lottery prizes in Norway. The second is US data from the Survey of Consumer Expectations, which asks respondents about their savings response to a change in income. The third is a new probability-based survey representing the US adult population that we conducted on the AmeriSpeak panel, in which we asked respondents about their savings behavior in response to a persistent raise in salary. These three sources are consistent in suggesting similar magnitudes of the causal effect of income on savings, with little variation across incomes or savings levels. Importantly, the causal income effect is meaningfully smaller than the slope of the across-income variation of savings—particularly at high incomes—implying an important role for savings taxes. Incorporated into our formulas, this implies a (mostly) positive and progressive optimal tax on savings. As in other work, these results are sensitive to the elasticity of savings with respect to tax rates, about which there is still substantial uncertainty. Accounting for multidimensional heterogeneity has a quantitatively modest impact on optimal tax rates in our calibration, whereas accounting for return heterogeneity has a larger impact.

Our paper contributes to a number of literatures. The first is the literature studying optimal commodity and savings taxation in extensions of the Atkinson and Stiglitz (1976) framework. Focusing on preference heterogeneity, Saez (2002) addresses the qualitative question of what observable statistics imply that a "small" *linear* commodity (savings) tax can increase welfare, but does not derive the optimal tax or address other forms of across-income heterogeneity. Golosov et al. (2013) derive conditions characterizing the optimal second-best allocation in a similar model but formulate their results in terms of first-order conditions on structural primitives rather than empirically estimable sufficient statistics.[4] Saez and Stantcheva (2018) study nonlinear capital taxation in a setting without income effects, which corresponds to the special case of our model where $s_{inc}'(z) = 0$. Scheuer and Slemrod (2020) derive a version of our Pareto efficiency condition for separable nonlinear tax systems in the special case of unidimensional heterogeneity in initial wealth. Allcott, Lockwood, and Taubinsky (2019) derive a sufficient statistics formula for the optimal separable linear commodity tax.[5] Hellwig and Werquin (2022) characterize optimal asymptotic savings and income tax rates at the top, in terms of elasticities and Pareto parameters of the tails of the income, wealth, and consumption distributions.[6] Looking at other sources of across-income heterogeneity, Gahvari and Micheletto (2016); Gerritsen et al. (2020); and Schulz (2021) study heterogeneous rates of return; Boadway, Marchand, and Pestieau (2000); Cremer, Pestieau, and Rochet (2003); and Scheuer and Slemrod (2020) study heterogeneous endowments; Christiansen and Tuomala (2008) study income shifting; and Bovenberg and Jacobs (2005) and Bovenberg and Jacobs (2011) study human capital investments.[7] Our results extend these insights by developing methods to characterize optimal taxes in terms of estimable sufficient statistics that hold for many of the deviations from the Atkinson-Stiglitz Theorem analyzed in prior work.

Second, we connect to the empirical literature measuring marginal propensities to consume, heterogeneous rates of return, and elasticities of wealth, savings, and capital income (e.g., Fagereng et al. 2020; Fagereng, Holm, and Natvik 2021; Agersnap and Zidar 2021; Jakobsen et al. 2020; Smith, Zidar, and Zwick 2021). Our formulas provide a direct link from these empirical statistics to optimal tax implications. Our formulas also clarify the importance of providing precise estimates of across-income heterogeneity in these empirical statistics.

Third, this paper complements the literature on dynamic taxation (see overviews by Golosov, Tsyvinski, and Werning 2006; Stantcheva 2020), which typically assumes homogeneous preferences, but derives a theoretically robust role for capital taxation via the inverse Euler equation (e.g., Golosov, Kocherlakota, and Tsyvinski 2003; Farhi and Werning 2013). Our work is complementary in relaxing the assumption of homogeneous and weakly separable preferences, but using a two-period framework. Quantitatively, the dynamic taxation literature tends to find optimal savings "wedges" of only several percentage points (see, e.g., Farhi and Werning 2013; Golosov and Tsyvinski 2015; Golosov, Troshkin, and Tsyvinski

[4] The empirical estimates in Golosov et al. (2013) suggest substantially less across-income heterogeneity than ours do, probably in part because they study heterogeneity in time discounting only, rather than the broader set of forces that can contribute to $s'(z) - s_{inc}'(z)$. This could also be driven by attenuation bias since they measure preference heterogeneity by regressing a structural estimate of time preferences on a plausibly noisy proxy of earnings ability (performance on the Armed Forces Qualification Test).

[5] The application of separable *linear* savings taxes in the presence of multidimensional heterogeneity is also considered in other work. Piketty and Saez (2013) derive sufficient statistics formulas but make the additional restriction of a linear income tax. Diamond and Spinnewijn (2011) and Gauthier and Henriet (2018) allow for a nonlinear income tax but assume a finite number of possible earnings levels and derive results in terms of model primitives. Jacquet and Lehmann (2021a) provide a generalization to a separable sum of many one-dimensional nonlinear tax schedules.

[6] They show that when individuals at the top do not adjust their consumption in response to changes in income but only adjust their wealth, the optimal top wealth tax rate can be nonzero, despite the fact that $s'(z) - s_{inc}'(z) = 0$. This is because the savings elasticity also converges to zero, while the ratio of these two statistics converges to a finite limit (see Hellwig and Werquin 2022).

[7] See Scheuer and Slemrod (2021) for a review of these motives for taxing wealth.

2016), substantially lower than those suggested by our baseline calibrations at the same assumed values of elasticities. This suggests that across-income heterogeneity may play a quantitatively larger role in determining optimal savings tax policy than do the social insurance motives analyzed in the dynamic taxation literature, and it motivates future research incorporating our method of measuring across-income heterogeneity into fully dynamic models.

The rest of this paper proceeds as follows. Section I presents a heuristic derivation of key formulas in a special case, to provide intuition for the main results. Section II introduces more formally our model and assumptions. Section III provides a sufficient statistics characterization of optimal smooth tax systems. Section IV shows how these sufficient statistics formulas span many different sources of across-income heterogeneity. Section V studies simple tax systems. Section VI presents extensions. Section VII applies our formulas to quantify optimal savings tax rates in the United States. Section VIII concludes. All proofs are gathered in the online Appendix.

I. Heuristic Derivation of a Special Case

In this section, we provide instructive and heuristic derivations for a special case of our general setting to distill the key ideas behind our main results. We formalize all of our assumptions and rigorously define all elasticity concepts in the subsequent sections that contain the general results.

A Sufficient Statistic for Preference Heterogeneity.—We study optimal taxation in a setting where individuals differ in their ability to earn labor income, z, and in their relative preference for two consumption goods, c and s, here interpreted as current consumption and savings. The government observes choices (c, s, z) and levies a nonlinear income tax, $T_z(z)$, and a nonlinear savings tax, $T_s(s)$. The individual budget constraint is $c + s = z - T_z(z) - T_s(s)$, with prices of c and s normalized to one.

To build intuition, in this section we consider the special case where savings are homogeneous conditional on earnings, so we can describe the observed profile of savings in the population as a function of earnings: $s(z)$. For concreteness, suppose that savings are increasing in earnings, $s'(z) > 0$, as illustrated in the left panel of Figure 1.

Our main contribution is to relate the optimal tax on s, $T_s(s)$, to an empirically measurable statistic, which we illustrate via Figure 1. Consider individuals at point A, with earnings z^0 and savings $s(z^0)$, and compare them to individuals at point D, with slightly higher earnings $z^0 + \Delta z$ and savings $s(z^0 + \Delta z)$. For a small value of Δz, the slope of the line connecting points A and D is approximately $s'(z^0)$, which captures the local cross-sectional variation in savings along the income distribution. Now suppose that individuals at point A increase their earnings by an amount Δz. Because of an income effect, they will also adjust their savings, for example, ending up at point C. Letting $s'_{inc}(z)$ denote the causal effect of income on savings, the slope of the line connecting points *A* and *C* will be approximately $s'_{inc}(z^0)$.

If points *C* and *D* coincide, then richer individuals choose higher savings solely due to the causal income effect of earnings on savings. This is the special case considered in the theorem of Atkinson and Stiglitz (1976), which states that redistribution

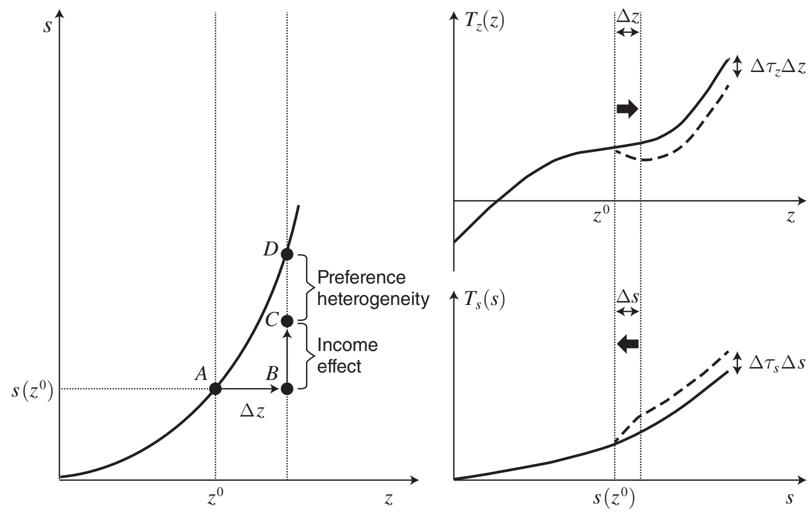

Figure 1. Intuitive Decomposition (Left Panel) and Joint Tax Reforms (Right Panel)

Notes: In the left panel, the line plots savings across the income distribution, together with its decomposition between income effects and other sources of heterogeneity, such as preferences. In the right panel, the solid lines represent income and savings tax schedules, and dashed lines represent the tax reforms that we consider. We set $\Delta s = s'(z^0)\Delta z$ so that both reform intervals contain the same set of individuals.

should be carried out only through a labor income tax, $T_z(z)$, so that the optimal savings tax $T_s(s)$ is everywhere zero.[8] In this case, a tax on savings discourages labor supply in the same way that a tax on earnings does. But it is less efficient because it additionally distorts savings choices conditional on a given choice of z.

If instead $C < D$ as illustrated in Figure 1, then richer individuals choose higher savings in part because they have, for example, a stronger preference for saving due to being more patient. In this case, the theorem of Atkinson and Stiglitz (1976) doesn't apply, and the optimal policy features a savings tax. The mechanism design literature often presents the intuition for this result in terms of information revelation: savings choices reveal information about underlying earning ability beyond what is revealed by earnings choices, which the mechanism designer can exploit using a savings tax.

In this paper, we move beyond the mechanism design approach by providing empirically grounded, quantitative formulas for optimal taxes on s. Specifically, we show that the optimal marginal savings tax rate is proportional to the gap between C and D, as measured by $s'(z) - s'_{inc}(z)$. This also provides alternative intuition for taxing s versus z in terms of their relative distortions to earnings because, as we explain below, the distortionary effects of the savings tax on earnings are proportional to $s'_{inc}(z)$. Empirically, this decomposition requires only cross-sectional data to measure the cross-sectional variation, $s'(z)$, and estimates of income effects by income groups, $s'_{inc}(z)$.

[8] Formally, the optimal savings tax can be any lump-sum amount, which we normalize to zero.

As a concrete example, suppose individuals have heterogeneous earnings ability θ so that to generate earnings z, an individual must exert effort of z/θ. They also have heterogeneous preferences for savings due to heterogeneous discount factors δ. The utility functions are given by

$$U(c,s,z;\delta,\theta) = u(c) + \delta v(s) - k(z/\theta), \tag{1}$$

where u and v are increasing concave functions and k is an increasing convex function representing disutility of effort. Let $S(z,\delta)$ denote the optimal choice of s by an individual earning z with a discount factor δ.[9] To microfound our assumption that there is a single value of s for each z, we assume that preference heterogeneity is unidimensional, meaning that we can write discount factors as a function of earning ability, $\delta(\theta)$. With minor abuse of notation, we can also write δ as a function of z instead of as a function of θ. This implies that we can write the slope of the across-income savings profile, $s'(z)$, as a sum of partial derivatives, which capture preference heterogeneity and income effects, respectively:

$$s'(z) = \underbrace{\frac{\partial S}{\partial z}}_{s'_{inc}(z)} + \underbrace{\frac{\partial S}{\partial \delta}\frac{\partial \delta}{\partial z}}_{\text{preference heterogeneity}}. \tag{2}$$

Thus, the preference heterogeneity statistic can be obtained from the difference $s'(z) - s'_{inc}(z)$.

Optimal Taxation of s.—We characterize optimal taxes on s by considering the effects of a joint reform of $T_z(z)$ and $T_s(s)$, illustrated in the right panel of Figure 1. First, consider a $\Delta\tau_z$ *decrease* in the marginal income tax rate in an earnings interval $[z^0, z^0 + \Delta z]$, where Δz is small and $\Delta\tau_z$ is much smaller. Second, consider a small $\Delta\tau_s$ *increase* in the marginal tax rate on s in the corresponding savings interval $[s^0, s^0 + \Delta s]$, where $s^0 = s(z^0)$ and $\Delta s = s'(z^0)\Delta z$. Because Δz is small, individuals with earnings $z^0 + \Delta z$ choose savings $s^0 + \Delta s$. Thus, there is just a single interval of individuals whose marginal tax rates on earnings and savings are altered.

We select $\Delta\tau_z$ and $\Delta\tau_s$ so that the labor supply responses to the income tax and savings tax reforms cancel out in this interval. Following Saez (2002), we show in Lemma 1 that for a z-earner, a small $\Delta\tau$ increase in the marginal tax rate on savings, $T'_s(s)$, induces the same earnings change as a $s'_{inc}(z)\Delta\tau$ increase in the marginal tax rate on earnings, $T'_z(z)$. To see the intuition, consider how each reform alters earning incentives. If the marginal tax rate on earnings z increases by $\Delta\tau_z$, a z-earner must pay $\Delta\tau_z$ more taxes on each additional dollar of earnings, so they reduce their earnings in response. Alternatively, if the marginal tax rate on savings s increases by $\Delta\tau_s$, and the individual raises s by s'_{inc} for every additional dollar of earnings, then they must pay $s'_{inc}\Delta\tau_s$ more taxes on each additional dollar of earnings, so they also reduce their earnings in response. The earnings responses are the same if $\Delta\tau_z = s'_{inc}(z)\Delta\tau_s$. Hence, our reform reduces income tax rates by $\Delta\tau_z = s'_{inc}(z^0)\Delta\tau_s$, so that earnings responses cancel out in this interval.

[9] Specifically, $S(z,\delta)$ satisfies the first-order condition $-u'\big(z - S(z,\delta) - T_z(z) - T_s(S(z,\delta))\big) + \delta v'\big(S(z,\delta)\big) = 0$.

As a result, the net effect of the joint reform is that individuals in this interval react only by reducing their savings in response to the increase in the marginal savings tax rate. This reduction is given by $ds = -\frac{s^0}{1 + T_s'(s^0)}\zeta_{s|z}^c(z^0)\Delta\tau_s$, where $\zeta_{s|z}^c(z^0) \geq 0$ denotes the compensated savings elasticity holding earnings fixed. Letting $h_z(z)$ denote the earnings density, the mass of individuals located in this interval is $h_z(z^0)\Delta z$. Thus, the overall revenue effect stemming from individuals in this interval is[10]

$$T_s'(s^0) \cdot ds \cdot h_z(z^0)\Delta z = T_s'(s^0) \cdot \left(-\frac{s^0}{1 + T_s'(s^0)}\zeta_{s|z}^c(z^0)\Delta\tau_s\right) \cdot h_z(z^0)\Delta z. \tag{3}$$

$$= -\frac{T_s'(s^0)}{1 + T_s'(s^0)} s^0 \zeta_{s|z}^c(z^0)\, h_z(z^0)\Delta\tau_s\Delta z.$$

Individuals with earnings $z < z^0$ are not affected by the reform. Individuals with earnings $z \geq z^0 + \Delta z$ experience a lump-sum change in their tax liability. The income tax reform generates a lump-sum tax decrease of $\Delta\tau_z\Delta z$, while the savings tax reform generates a lump-sum tax increase of $\Delta\tau_s s'(z^0)\Delta z$. Thus, the net change in tax liability is

$$\begin{aligned}\Delta T &= \Delta\tau_s s'(z^0)\Delta z - \Delta\tau_z\Delta z \\ &= \Delta\tau_s s'(z^0)\Delta z - \left[s_{inc}'(z^0)\Delta\tau_s\right]\Delta z \\ &= \left[s'(z^0) - s_{inc}'(z^0)\right]\Delta\tau_s\Delta z.\end{aligned}$$

Let $\hat{g}(z)$ denote the social value, in units of public funds, of giving a dollar to an individual with earnings z, including any fiscal externalities from income effects. Then the net revenue and welfare effect of the joint reform from individuals with earnings $z \geq z^0 + \Delta z$ is

$$\Delta T \cdot \int_{z \geq z^0 + \Delta z}\left[1 - \hat{g}(z)\right] h_z(z)\, dz. \tag{4}$$

The optimal tax system is one that cannot be improved upon, implying that the effects in (3) and (4) must sum to zero. Rearranging, and taking the limit $\Delta z \to 0$, yields the optimal savings tax condition:

$$\frac{T_s'(s^0)}{1 + T_s'(s^0)} = \left[s'(z^0) - s_{inc}'(z^0)\right]\frac{1}{s^0\zeta_{s|z}^c(z^0)}\frac{1}{h_z(z^0)}\int_{z \geq z^0}\left[1 - \hat{g}(z)\right] h_z(z)dz. \tag{5}$$

[10] Note that the envelope theorem implies that there is no (first-order) welfare effect in this interval.

Formula (5) shows that optimal marginal tax rates on s satisfy a condition that is remarkably similar to the formula for optimal marginal tax rates on z derived in Diamond (1998) and Saez (2001). Importantly, the formula provides a transparent generalization of the Atkinson-Stiglitz Theorem. When $s'(z) - s'_{inc}(z)$ is equal to zero, the formula implies that the optimal tax on s must equal zero as well. When $s'(z) - s'_{inc}(z)$ is positive, implying that higher earners have a stronger relative preference for s, the optimal tax rate on s is positive. Intuitively, the higher is $s'_{inc}(z^0)$, the more savings taxes depress earnings z, and thus, the less attractive savings taxes are. Thus, our formulas provide a counterpart to the mechanism design intuition by expressing optimal tax rates on savings in terms of their distortionary effects.

Equation (5) also quantifies several other forces governing optimal taxation of savings. Marginal tax rates on savings tend to be decreasing in the compensated elasticity of savings, $\zeta^c_{s|z}(z^0)$; decreasing in the mass of individuals being affected, $h_z(z^0)$; and increasing in the social value of a lump-sum tax on individuals with higher earnings, $\int_{z \geq z^0} [1 - \hat{g}(z)]\, h_z(z)\, dz$.

Pareto Efficiency Conditions.—Because certain earnings tax reforms may be more or less distortionary than certain savings tax reforms, not all combinations of savings and earnings taxes are Pareto efficient. Our techniques also allow us to characterize Pareto-efficient tax systems.

To do so, instead of setting $\Delta\tau_z = s'_{inc}(z^0)\Delta\tau_s$ to cancel out changes in earnings for individuals located in the reform interval, we set $\Delta\tau_z = s'(z^0)\Delta\tau_s$ to cancel out changes in tax *liability* for individuals located *above* the reform interval. For individuals located in the reform interval, this joint reform then produces a net earnings change of $dz = \left[s'(z^0) - s'_{inc}(z^0)\right] \frac{z^0}{1 - T'_z(z^0)} \zeta^c_z(z^0)\Delta\tau_s$, where ζ^c_z denotes the compensated elasticity of taxable income. This earnings change affects income tax revenue in proportion to the marginal income tax rate, $T'_z(z^0)$, and it also induces an indirect adjustment in savings of $s'_{inc}(z)dz$ that in turn affects savings tax revenue in proportion to the marginal savings tax rate, $T'_s(s^0)$. Earnings changes thus generate a fiscal externality that changes tax revenue by $\left[T'_z(z^0) + s'_{inc}(z^0)\, T'_s(s^0)\right] \cdot dz \cdot h_z(z^0)\Delta z$.

Moreover, for individuals located in the reform interval, the increase in marginal savings tax rates also induces a direct adjustment in savings of $ds = -\frac{s^0}{1 + T'_s(s^0)} \zeta^c_{s|z}(z^0)\Delta\tau_s$, generating an additional fiscal externality that alters tax revenue by $T'_s(s^0) \cdot ds \cdot h_z(z^0)\Delta z$.

Pareto efficiency requires an absence of Pareto-improving reforms, meaning that the two fiscal externalities must sum to zero. Rearranging this condition implies that a Pareto-efficient tax system satisfies

$$(6) \quad \frac{T'_s(s^0)}{1 + T'_s(s^0)} = \left[s'(z^0) - s'_{inc}(z^0)\right] \frac{z^0 \zeta^c_z(z^0)}{s^0 \zeta^c_{s|z}(z^0)} \frac{T'_z(z^0) + s'_{inc}(z^0)\, T'_s(s^0)}{1 - T'_z(z^0)}.$$

This formula shows that any Pareto-efficient tax system must feature nonzero tax rates on s when $s'(z^0) - s'_{inc}(z^0) \neq 0$, meaning that there is across-income heterogeneity in preferences or other factors that we discuss in Section IV. When $s'(z^0) - s'_{inc}(z^0) > 0$, a higher ratio of the elasticity of earnings to that of savings, $\zeta_z^c(z^0)/\zeta_{s|z}^c(z^0)$, implies that it is more efficient to tax s instead of z.

The simplicity of this formula, and the fact that it does not depend on welfare weights, makes it particularly amenable for empirical applications. In Section VII, for example, we compute the Pareto-efficient savings tax given the existing income tax, without making any assumptions about redistributive motives.

Multidimensional Heterogeneity.—The derivations above assume that there is a single level of savings at each level of earnings. In much of the paper, we also consider multidimensional heterogeneity, by which we mean heterogeneity in savings at each earnings level. In this more general case, one candidate for the analog to $s'\left(z^0\right)$ would be $\frac{d}{dz^0}\left(\mathbb{E}\left[s|z^0\right]\right)$, the change in average savings with income, and correspondingly, the analog to $s'_{inc}\left(z^0\right)$ would be $\mathbb{E}\left[s'_{inc}|z^0\right]$, the average causal income effect on savings of all z^0-earners. These are the appropriate analogs for some of the simple tax systems that we consider in Section V. However, when we consider fully flexible tax systems where conditional on earnings, the marginal tax rate on savings can still vary with s, we use more general statistics that involve *conditional* averages above a given s^0 : $\frac{d}{dz^0}\left(\mathbb{E}\left[s\,|\,s \geq s^0, z^0\right]\right)$ and $\mathbb{E}\left[s'_{inc}\,|\,s \geq s^0, z^0\right]$.

II. Baseline Model and Assumptions

We now present the more general model on which the results in the remainder of the paper are based. For concreteness, we begin by discussing preference heterogeneity and then generalize the analysis to broader forms of across-income heterogeneity in Section IV.

Setting.—There is a population of heterogeneous individuals indexed by their type $\theta \in \Theta$, where Θ is compact. They differ in their disutility from generating earnings z and in their preferences for a consumption bundle $\left(c,s\right)$ embodied in their utility function $U\left(c,s,z;\theta\right)$. Heterogeneity is unidimensional when $\Theta \subset \mathbb{R}$, in which case we interpret type θ as earnings ability, and other dimensions of heterogeneity can be written as functions of θ. Heterogeneity is multidimensional when $\Theta \subset \mathbb{R}^n$, in which case earnings ability, $w\left(\theta\right)$, may vary independently of other dimensions of heterogeneity. We assume that θ has a differentiable cumulative distribution function $F\left(\theta\right)$.

One application is where c is period 1 consumption and s is the realized savings consumed in period 2, as in Saez (2002); Golosov et al. (2013); and many others. A second application is where c is period 1 consumption by the parents, while s is the wealth bequeathed to their children and consumed in period 2 (e.g., Farhi and Werning 2010). A third application is where c is numeraire consumption and s is a particular commodity that could be taxed nonlinearly, such as energy or housing (e.g., Gaubert, Kline, and Yagan 2021).

Throughout the paper, we assume the following regularity conditions for the utility function.

ASSUMPTION 1: *$U(c,s,z;\theta)$ is twice continuously differentiable, increasing and weakly concave in c and s, and decreasing and strictly concave in z. The first derivatives U_c' and U_s' are bounded.*

We also assume a linear production technology with marginal rate of transformation p between s and c, an assumption that we later generalize. In the savings and inheritance interpretations of the model, $p = 1/R$, where R is the gross rate of return in a linear savings technology between the two periods.

Atkinson and Stiglitz (1976) assume that preferences are weakly separable between the utility from consumption and the disutility from labor and that preferences for consumption are homogeneous, e.g., $U(c,s,z;\theta) = u(c,s) - k(z/\theta)$, with $u(.)$ the utility from consumption and $k(.)$ the disutility from work. In contrast, Assumption 1 allows for failures of weak separability between consumption and labor, as in Corlett and Hague (1953). It also allows for preference heterogeneity, e.g., $U(c,s,z;\theta) = u(c) + \delta(\theta)u(s) - k(z/\theta)$, where $\delta(\theta)$ represents heterogeneous discount factors as in example (1).

Government.—The government maximizes a weighted sum of utilities,

$$\max \int_\theta \alpha(\theta) U(c(\theta), s(\theta), z(\theta); \theta) dF(\theta), \tag{7}$$

where $\alpha(\theta)$ are Pareto weights. Selecting a particular set of weights requires normative assumptions, which we discuss when introducing social marginal welfare weights in Section IIIA.

Type θ is private information and cannot be observed by the government; only the distribution of types, $F(\theta)$, is known. Therefore, the government designs a tax and transfer function $\mathcal{T}$ that depends on the observable variables c, s, and z, which can be written without loss of generality as a tax on s and z only, $\mathcal{T}(s,z)$.[11] The government anticipates that individuals choose these variables to maximize their utility subject to their individual budget constraints, $c + ps \leq z - \mathcal{T}(s,z)$. Thus, an optimal tax system maximizes the objective in (7) subject to individual optimization and subject to a resource constraint, $\int_\theta \mathcal{T}(s(\theta), z(\theta)) dF(\theta) \geq E$, where E is an exogenous revenue requirement.

If the tax system $\mathcal{T}(s,z)$ is unrestricted, this problem is equivalent to the problem of selecting an optimal allocation $\mathcal{A} = \{(c(\theta), s(\theta), z(\theta))\}_\theta$ to maximize the objective in (7) subject to the resource constraint

$$\int_\theta [z(\theta) - ps(\theta) - c(\theta)] \, dF(\theta) \geq E \tag{8}$$

[11] Expressing the tax function as $\mathcal{T}(c,s,z)$ is redundant. Indeed, any choice of s and z then implies a consumption value given by $\mathcal{C}(s,z) := \max\{c | c = z - s - \mathcal{T}(c,s,z)\}$, implying that one can reexpress the tax as $\tilde{\mathcal{T}}(s,z) = \mathcal{T}(\mathcal{C}(s,z), s, z)$.

and subject to incentive compatibility constraints

$$(9) \qquad \forall (\theta, \theta') \in \Theta^2, \; U(c(\theta), s(\theta), z(\theta); \theta) \geq U(c(\theta'), s(\theta'), z(\theta'); \theta).$$

We refer to an allocation $\mathcal{A} = \{(c(\theta), s(\theta), z(\theta))\}_\theta$ that maximizes (7) subject to (8) and (9) as an *optimal incentive-compatible allocation*.

When heterogeneity is unidimensional, we show in online Appendix D.A that under regularity conditions (Assumption A1), an optimal incentive-compatible allocation can be implemented by a *smooth* tax system (Theorem A1), that is, by twice continuously differentiable tax functions $\mathcal{T}(s,z)$. This motivates our focus on smooth tax systems in the rest of the paper.

III. Optimal Smooth Tax Systems

In this section we formally define all relevant sufficient statistics. We then provide a formal characterization of optimal smooth tax systems under unidimensional heterogeneity and multidimensional heterogeneity.

A. *Sufficient Statistics for Smooth Tax Systems*

Assumptions: To define our sufficient statistics, it is helpful to write an individual's optimization problem under a tax system $\mathcal{T}(s,z)$ as

$$(10) \qquad \max_z \left\{ \max_{c,s} U(c,s,z;\theta) \quad \text{subject to} \quad c \leq z - ps - \mathcal{T}(s,z) \right\},$$

where the inner problem represents the optimal choices of $c(z,\theta)$ and $s(z,\theta)$ for a given earnings level z and the outer problem represents the optimal choice of earnings $z(\theta)$ taking into account endogenous choices of c and s. Our results concern tax systems that satisfy the following conditions.

CONDITION 1: *The tax system $\mathcal{T}$ (i) is twice differentiable in s and z, (ii) induces an allocation where c, s, and z are smooth functions of θ, and (iii) has unique individual optima where second-order conditions strictly hold.*

These regularity assumptions allow the use of variational methods to characterize optimal tax systems. Together with the implicit function theorem, they ensure that individuals' chosen allocations will vary smoothly with tax reforms in a neighborhood of that optimal tax system.[12]

[12] Golosov and Krasikov (2023) show that conditions like these are plausible even with multidimensional heterogeneity. This is in contrast to monopolist screening problems, where participation constraints bind for many types, leading to nonsmooth responses. With optimal tax systems, the first-order approach is generally valid as long as welfare weights do not vary too sharply with types or are not too redistributive. Intuitively, when welfare weights are homogeneous and there are no income effects, the optimal tax is uniform and thus smooth, implying that moderate deviations from this case still leave the optimal tax system smooth. See Bergstrom and Dodds (2021) or Dodds (2023) for recent work on optimal taxation when these conditions are not satisfied.

CONDITION 2: *We also impose one of two additional conditions on the tax systems* $\mathcal{T}$ *that we consider:*

(UD) $\Theta \subset \mathbb{R}$ *and in the allocation induced by* $\mathcal{T}$, *any two types* $\theta \neq \theta'$ *choose different earnings*, $z(\theta) \neq z(\theta')$;

(MD) $\Theta \subset \mathbb{R}^n$ *and in the allocation induced by* $\mathcal{T}$, *there are no atoms in the distribution of* (c, s, z).

Condition (UD) generates what we refer to as unidimensional heterogeneity; it ensures that c and s can be represented as functions of z. In other words, this assumption rules out the possibility that individuals with the same earnings choose different consumption bundles.

Condition (MD) concerns the multidimensional heterogeneity case, where there can be many different bundles of (c, s) for any given z. It is most natural to apply this condition when $\Theta \subset \mathbb{R}^n$ for $n \geq 2$, though our results also apply to cases where this condition holds and $\Theta \subset \mathbb{R}$.

To span both unidimensional and multidimensional cases, we define type-specific sufficient statistics. With unidimensional heterogeneity, sufficient statistics can be equivalently expressed as functions of earnings, z. With multidimensional heterogeneity, sufficient statistics are generally heterogeneous within earnings, and we express optimal tax formulas using averages of type-specific sufficient statistics. We use $H(s, z)$ and $h(s, z)$ to denote the cumulative distribution and density functions over (s, z), with h_s and h_z denoting the marginal densities over s and z.

Elasticities for z-choices: Earnings responses to tax reforms are quantified by ζ_z^c, the compensated elasticity of labor income with respect to the marginal earnings tax rate, and η_z, the labor income effect parameter. Formally, for each type θ, using the shorthand $s(\theta) := s(z(\theta), \theta)$ to economize on notation, we define

$$\zeta_z^c(\theta) := -\frac{1 - \mathcal{T}_z'(s(\theta), z(\theta))}{z(\theta)} \frac{\partial z(\theta)}{\partial \mathcal{T}_z'(s(\theta), z(\theta))} \tag{11}$$

$$\eta_z(\theta) := -\left[1 - \mathcal{T}_z'(s(\theta), z(\theta))\right] \frac{\partial z(\theta)}{\partial \mathcal{T}(s(\theta), z(\theta))}, \tag{12}$$

where $\mathcal{T}(s(\theta), z(\theta))$ is the tax liability and $\mathcal{T}_z'(s(\theta), z(\theta))$ is the marginal labor income tax rate. Since earnings choices take into account endogenous choices of c and s, these elasticities include the full sequence of adjustments due to changes in choices of c and s, as well as those due to nonlinearities in the tax system.[13]

[13]This corresponds to the circular adjustments described in, e.g., Jacquet, Lehmann, and Van der Linden (2013); Golosov, Tsyvinski, and Werquin (2014); Scheuer and Werning (2017); Jacquet and Lehmann (2021b).

Elasticities for *s*-choices: Changes in s in response to tax reforms are quantified by $\zeta^c_{s|z}$, the compensated elasticity of s with respect to the marginal tax rate on s, $\eta_{s|z}$, the income effect parameter, and s'_{inc}, the causal effect on consumption of s from a marginal change in earnings z. Formally,

$$\zeta^c_{s|z}(\theta) := -\frac{1+\mathcal{T}'_s(s(z,\theta),z)}{s(z,\theta)} \frac{\partial s(z,\theta)}{\partial \mathcal{T}'_s(s(z,\theta),z)}\bigg|_{z=z(\theta)} \tag{13}$$

$$\eta_{s|z}(\theta) := -\left[1+\mathcal{T}'_s(s(z,\theta),z)\right]\frac{\partial s(z,\theta)}{\partial \mathcal{T}(s(z,\theta),z)}\bigg|_{z=z(\theta)} \tag{14}$$

$$s'_{inc}(\theta) := \frac{\partial s(z,\theta)}{\partial z}\bigg|_{z=z(\theta)}, \tag{15}$$

where $\mathcal{T}'_s(s(z,\theta),z)$ is the marginal tax rate on s of a type θ who earns labor income z. The parameters $\zeta^c_{s|z}$ and $\eta_{s|z}$ measure responses to tax reforms accounting for nonlinearities in the tax system, holding labor income fixed. In particular, the income effect parameter $\eta_{s|z}$ measures changes in s due to windfall changes in disposable income. In contrast, s'_{inc} measures changes in s that are induced by changes in earnings. Hence, the parameters $\eta_{s|z}$ and s'_{inc} both measure causal income effects on savings, of after-tax windfalls and changes in gross earnings, respectively. If preferences are weakly separable between consumption and labor and the tax system is separable between s and z, then $s'_{inc}(\theta) = \frac{1-\mathcal{T}'_z}{1+\mathcal{T}'_s}\eta_{s|z}(\theta)$ (see Proposition 2).

Note that we define the elasticity of s with respect to $1+\mathcal{T}'_s$ rather than $p+\mathcal{T}'_s$. This choice is irrelevant when $p \equiv 1$—a normalization that can be adopted without loss of generality, as shown in our discussion of generalized budget constraints in Section IVA. However, defining the elasticity with respect to $p+\mathcal{T}'_s$ may be more natural in applications where s is a commodity sold at an after-tax price of $q = p+\mathcal{T}'_s$.[14]

Social Marginal Welfare Weights.—To encode the policymaker's redistributive tastes, we follow the literature in defining social marginal welfare weights as the change in social welfare from a marginal increase in consumption for an individual of type θ, normalized by the marginal value of public funds λ:[15]

$$g(\theta) := \frac{\alpha(\theta)}{\lambda}U'_c\big(z(\theta) - \mathcal{T}(s(\theta),z(\theta)) - ps(\theta), s(\theta), z(\theta);\theta\big). \tag{16}$$

[14] It is straightforward to convert our results between these elasticity definitions: in this case, the appropriate formulas can be obtained from Theorem 1 and Proposition 3 by multiplying $\zeta^c_{s|z}$ by $(p+\mathcal{T}'_s)/(1+\mathcal{T}'_s)$. The only change in Theorem 1 is that the left-hand side in equation (19) becomes $\frac{\mathcal{T}'_s(s(z),z)}{p+\mathcal{T}'_s(s(z),z)}$, and analogously for Proposition 3.

[15] These welfare weights should be understood as arising from the weighted utilitarian social objective in equation (7), analogous to β^h in Diamond (1975) and Atkinson and Stiglitz (1976). Saez and Stantcheva (2016) propose extending such weights to nonutilitarian normative objectives, although Sher (2023) presents limitations of that approach.

We define $\hat{g}(\theta)$ as the social marginal welfare weights augmented with the fiscal impact of income effects as in, e.g., Diamond (1975). This represents the full social value of marginally increasing the disposable income of a θ-type individual. Formally,

$$\hat{g}(\theta) := g(\theta) + \frac{\mathcal{T}_z'\big(s(\theta),z(\theta)\big) + s_{inc}'(\theta)\,\mathcal{T}_s'\big(s(\theta),z(\theta)\big)}{1 - \mathcal{T}_z'\big(s(\theta),z(\theta)\big)}\eta_z(\theta) + \frac{\mathcal{T}_s'\big(s(\theta),z(\theta)\big)}{1 + \mathcal{T}_s'\big(s(\theta),z(\theta)\big)}\eta_{s|z}(\theta). \tag{17}$$

The term proportional to η_z captures fiscal externalities arising from changes in earnings choices (including the indirect changes in s, proportional to s_{inc}'). The term proportional to $\eta_{s|z}$ captures the fiscal externalities arising from direct changes in s induced by changes in disposable income.

Social marginal welfare weights embed judgments about interpersonal utility comparisons. These are usually treated as normative assumptions, although some research has utilized survey data to estimate these weights (see Appendix C of Saez and Stantcheva 2016) or estimated them from existing policies via an "inverse optimum" procedure (e.g., Bourguignon and Spadaro 2012; Lockwood and Weinzierl 2016). Such normative assumptions are particularly strong when there is preference heterogeneity because individuals prefer different bundles—and face different tax burdens—even when they have identical budget sets. For example, a savings tax might be viewed as an unfair tax on those with relatively high discount factors. Lockwood and Weinzierl (2015) show that this difficulty arises even in the standard Mirrlees (1971) model because there is no empirical distinction between heterogeneous earnings ability and heterogeneous preferences for exerting labor effort. We write our theoretical results in terms of flexible welfare weights that span the degree of heterogeneity in individuals' types so that optimal policy can be computed using whatever welfare weights the policymaker prefers.[16]

B. *Characterization of Optimal Smooth Tax Systems*

Unidimensional Heterogeneity.—As discussed in Section I, the statistic s_{inc}' relates how earnings respond to reforms to marginal tax rates on s. We formalize this below in a lemma that generalizes Lemma 1 in Saez (2002).

LEMMA 1: *A small increase $d\tau$ in the marginal tax rate on s faced by an individual earning z induces the same earnings change as a small increase $s_{inc}'(z)\,d\tau$ in the marginal tax rate on z.*

When types are unidimensional, we obtain the following characterization of optimal smooth tax systems, which generalizes the heuristic derivation in Section I.

[16] Our empirical application in Section VII employs a version of the inverse optimum approach, estimating optimal savings taxes consistent with the current US taxes on labor income.

THEOREM 1: *Consider an optimal tax system for which Conditions 1 and 2-UD hold. At each bundle $\left(c^0, s^0, z^0\right)$ chosen by a type θ, this tax system satisfies*

$$(18)\quad \frac{\mathcal{T}_z'\left(s^0, z^0\right)}{1 - \mathcal{T}_z'\left(s^0, z^0\right)} = \frac{1}{z^0 \zeta_z^c\left(z^0\right)} \frac{1}{h_z\left(z^0\right)} \int_{z \geq z^0} \left[1 - \hat{g}(z)\right] dH_z(z) - s_{inc}'\left(z^0\right) \frac{\mathcal{T}_s'\left(s^0, z^0\right)}{1 - \mathcal{T}_z'\left(s^0, z^0\right)},$$

$$(19)\quad \frac{\mathcal{T}_s'\left(s^0, z^0\right)}{1 + \mathcal{T}_s'\left(s^0, z^0\right)} = \left[s'\left(z^0\right) - s_{inc}'\left(z^0\right)\right] \frac{1}{s^0 \zeta_{s|z}^c\left(z^0\right)} \frac{1}{h_z\left(z^0\right)} \int_{z \geq z^0} \left[1 - \hat{g}(z)\right] dH_z(z).$$

Consider a Pareto-efficient tax system for which Conditions 1 and 2-UD hold. At each bundle $\left(c^0, s^0, z^0\right)$ chosen by a type θ, this tax system satisfies

$$(20)\quad \frac{\mathcal{T}_s'\left(s^0, z^0\right)}{1 + \mathcal{T}_s'\left(s^0, z^0\right)} = \left[s'\left(z^0\right) - s_{inc}'\left(z^0\right)\right] \frac{z^0 \zeta_z^c\left(z^0\right)}{s^0 \zeta_{s|z}^c\left(z^0\right)} \frac{\mathcal{T}_z'\left(s^0, z^0\right) + s_{inc}'\left(z^0\right) \mathcal{T}_s'\left(s^0, z^0\right)}{1 - \mathcal{T}_z'\left(s^0, z^0\right)}.$$

Formula (18) for optimal marginal tax rates on z is analogous to the main result in Saez (2001), except for the last term. This term accounts for the fact that changes in earnings lead to adjustments in consumption of s that are proportional to $s_{inc}'(z)$ and therefore affect tax revenue when tax rates on s are nonzero. For "normal" goods ($s_{inc}'(z) > 0$), this additional distortion pushes for lower (respectively, higher) marginal tax rates on earnings, provided that the marginal tax rate on s is positive (respectively, negative).

Formula (19) is a key result that generalizes the heuristic derivation in Section I. Optimal marginal tax rates on s satisfy a condition that is remarkably similar to the standard formula for optimal marginal income tax rates, and that provides a transparent generalization of the Atkinson-Stiglitz Theorem. The optimal tax rate on s is zero, positive, or negative, respectively, when $s'(z) - s_{inc}'(z)$ is zero, positive, or negative. The magnitude of the tax rate on s tends to be decreasing in the elasticity of s with respect to the tax rate, increasing in the strength of redistributive motives, and decreasing in the density of individuals at point $s(z)$.[17]

Formula (20) is a Pareto efficiency condition, which can be obtained for optimal tax systems by combining formulas (18) and (19).[18] It quantifies the efficient balance between taxing s and taxing z, given the measure $s'(z) - s_{inc}'(z)$ of how relative

[17] As is standard for first-order conditions of this kind, such equations provide guidance about the forces governing optimal taxes, but one should use caution when formally interpreting comparative statics because the statistics in the formula are endogenous to the tax function itself.

[18] This result extends Konishi (1995); Laroque (2005); and Kaplow (2006), who derive Pareto efficiency conditions under the more restrictive assumptions of the Atkinson-Stiglitz Theorem.

tastes for s covary with earnings ability. The stronger the association between relative preferences for s and earnings ability, the more efficient it is to tax s instead of z. An important implication of this Pareto efficiency condition is that any Pareto-efficient tax system must feature nonzero tax rates on s when there is across-income heterogeneity in preferences or other factors that we discuss in Section IV.

Multidimensional Heterogeneity.—When types θ are multidimensional, there is heterogeneity in s conditional on z, and multiple types θ may choose the same allocation (s,z). For any two variables $X(\theta), Y(\theta)$, we use the shorthand notation

$$\mathbb{E}\left[X \middle| s \geq s^0, z^0\right] := \mathbb{E}\left[X(\theta) \middle| s(\theta) \geq s^0, z(\theta) = z^0\right], \tag{21}$$

$$\operatorname{cov}\left[X, Y \middle| s \geq s^0, z^0\right] := \operatorname{cov}\left[X(\theta), Y(\theta) \middle| s(\theta) \geq s^0, z(\theta) = z^0\right], \tag{22}$$

to denote averages and covariances among individuals with $s \geq s^0$ and earnings $z = z^0$. We also introduce the notation $\tilde{h}(s^0, z^0) := \left[1 - H_{s|z}(s^0|z^0)\right] h_z(z^0)$, which is the measure of individuals with $s \geq s^0$ and earnings $z = z^0$. Finally, we denote the fiscal externality from earnings substitution effects in response to changes in marginal tax rates on z,

$$FE_z(\theta) := \frac{\mathcal{T}_z'(s(\theta), z(\theta)) + s_{inc}'(\theta)\, \mathcal{T}_s'(s(\theta), z(\theta))}{1 - \mathcal{T}_z'(s(\theta), z(\theta))} z(\theta)\, \zeta_z^c(\theta). \tag{23}$$

Using this notation, we derive the following conditions characterizing optimal smooth tax systems with multidimensional heterogeneity, suppressing some arguments for simplicity. To simplify exposition, and to connect more tightly to our unidimensional results, we express our main results on optimal marginal tax rates on s in terms of the average marginal tax rate on individuals with earnings z^0 and savings above s^0.

THEOREM 2: *Consider an optimal tax system for which Conditions 1 and 2-UD hold. At each bundle (c^0, s^0, z^0) chosen by types θ, this tax system satisfies*

$$\mathbb{E}\left[FE_z \middle| s \geq s^0, z^0\right] = \frac{1}{\tilde{h}(s^0, z^0)} \int_{z \geq z^0} \mathbb{E}\left[1 - \hat{g} \,|\, s \geq s^0, z\right] \tilde{h}(s^0, z)\, dz, \tag{24}$$

$$\begin{aligned}
&\mathbb{E}\left[\frac{\mathcal{T}_s'}{1 + \mathcal{T}_s'} s\, \zeta_{s|z}^c \middle| s \geq s^0, z^0\right] \\
&= \left[\frac{d}{dz^0}\left(\mathbb{E}\left[s \,|\, s \geq s^0, z^0\right]\right) - \mathbb{E}\left[s_{inc}' \,|\, s \geq s^0, z^0\right]\right] \\
&\quad \times \frac{1}{\tilde{h}(s^0, z^0)} \int_{z \geq z^0} \mathbb{E}\left[1 - \hat{g} \,|\, s \geq s^0, z\right] \tilde{h}(s^0, z)\, dz \\
&\quad - \operatorname{cov}\left[\hat{g}, s \,|\, s \geq s^0, z^0\right] - \operatorname{cov}\left[FE_z, s_{inc}' \,|\, s \geq s^0, z^0\right] \\
&\quad + \frac{1}{\tilde{h}(s^0, z^0)} \frac{d}{dz^0}\left[\operatorname{cov}\left[FE_z, s \,|\, s \geq s^0, z^0\right] \tilde{h}(s^0, z^0)\right].
\end{aligned} \tag{25}$$

Theorem 2 generalizes Theorem 1 to settings with multidimensional heterogeneity. To see this, first note that formula (24) for optimal marginal tax rates on z reduces to a rearrangement of formula (18) in the unidimensional case, if we set $s^0 = s(z^0)$ such that $\tilde{h}(s^0, z^0) = h_z(z^0)$ and

$$\mathbb{E}\left[FE_z \mid s \geq s^0, z^0\right] = \frac{\mathcal{T}_z'(s^0, z^0) + s_{inc}'(z^0)\,\mathcal{T}_s'(s^0, z^0)}{1 - \mathcal{T}_z'(s^0, z^0)} z^0 \zeta_z^c(z^0).$$

Formula (25) for optimal marginal tax rates on s extends formula (19). In the second line, $s'(z) - s_{inc}'(z)$ is replaced by its multidimensional counterpart, $\frac{d}{dz^0}\left(\mathbb{E}\left[s \mid s \geq s^0, z^0\right]\right) - \mathbb{E}\left[s_{inc}' \mid s \geq s^0, z^0\right]$. Intuitively, the term $\frac{d}{dz^0}\left(\mathbb{E}\left[s \mid s \geq s^0, z^0\right]\right)$—which captures how average savings above s^0 vary along the income distribution—is the analog to $s'(z)$. Using *conditional* averages of savings is necessary in the multidimensional case because savings s are heterogeneous at each earnings level.

The final two lines of (25) feature additional covariance terms, new to this multidimensional setting. The first covariance term arises because lower redistributive motives toward those with higher s above s^0 (negative covariance) imply a higher optimal tax burden on those individuals and thus higher marginal tax rates on s. The second covariance term arises because higher marginal tax rates on s distort choices of z in proportion to s_{inc}' (Lemma 1). Therefore, a higher covariance between s_{inc}' and the fiscal externality from labor supply distortions makes taxing s more distortionary and leads to lower marginal tax rates on s.

The logic underlying the final covariance term is nuanced and relates to how marginal tax rates on s should vary optimally with z. Consider a reform that for all $s \geq s^0$ increases $\mathcal{T}_s'$ at z^0 by raising $\frac{d}{dz}\mathcal{T}_s'$ a little to the left of z and reducing $\frac{d}{dz}\mathcal{T}_s'$ a little to the right of z by a small constant amount η, leaving $\mathcal{T}_s'$ unchanged outside of the small neighborhood around z^0.[19] To the left of z^0, this reform *increases* the marginal tax rate on earnings, to a degree that is proportional to $s - s^0$, reducing labor supply and resulting in a negative fiscal externality proportional to $\text{cov}\left[FE_z, s \mid s \geq s^0, z\right]\tilde{h}(s^0, z)$. Conversely, to the right of z^0, this reform *reduces* the marginal tax rate on earnings, raising labor supply and producing a positive fiscal externality proportional to $\text{cov}\left[FE_z, s \mid s \geq s^0, z\right]\tilde{h}(s^0, z)$. The total impact of this reform depends on whether $\text{cov}\left[FE_z, s \mid s \geq s^0, z\right]\tilde{h}(s^0, z)$ is higher for z just above or below z^0—that is, it depends on how this covariance *changes with* z at z^0. When $\frac{d}{dz^0}\left[\text{cov}\left[FE_z, s \mid s \geq s^0, z^0\right]\tilde{h}(s^0, z^0)\right]$ is positive, this effect is fiscally beneficial, pushing toward a higher optimal $\mathcal{T}_s'$.

[19] Specifically, for some small ε and $\underline{z} = z^0 - \varepsilon$, $\overline{z} = z^0 + \varepsilon$, consider changing the tax burden by an amount equal to $(z - \underline{z})(s - s^0)\eta$ for $z \in [\underline{z}, z^0]$ and $s \geq s^0$, and by $(\overline{z} - z)(s - s^0)\eta$ for $z \in [z^0, \overline{z}]$ and $s \geq s^0$, where η is much smaller than ε. Taking the derivative with respect to z shows that the change in the marginal tax rate on earnings will be linear in s above s^0.

Overall, these results clarify when optimal taxes under multidimensional heterogeneity substantially differ from those under unidimensional heterogeneity. If most of the variation in sufficient statistics is across, but not within, income levels z, then unidimensional tax formulas lead to policies that are close to being optimal. In contrast, if there is a large variation in sufficient statistics within income levels z, this is unlikely to be the case, and our multidimensional tax formulas can be used to quantify these deviations.[20] A conceptual difference is that we can no longer combine optimal tax formulas to obtain Pareto efficiency conditions under multidimensional heterogeneity: tax reforms inevitably lead to vertical or horizontal redistribution.

IV. Across-Income Heterogeneity and Its Determinants

This section shows that our sufficient statistic formulas encompass many different forms of across-income heterogeneity besides preferences.

A. *Budget Heterogeneity and Auxiliary Choices*

So far, we have considered economies with type-specific preferences $U(c,s,z;\theta)$ but type-independent budget constraints $c \leq B(s,z) - \mathcal{T}(s,z)$, with budget domain $B(s,z) := z - ps$. In this environment, across-income heterogeneity captured by the sufficient statistic $s'(z) - s'_{inc}(z)$, or by its multidimensional analog, originates from preference heterogeneity only.

Our approach extends to other sources of across-income heterogeneity. For example, across-income heterogeneity in prices of s (Gahvari and Micheletto 2016; Gerritsen et al. 2020), in income shifting (Slemrod 1995; Christiansen and Tuomala 2008), and in endowments (Boadway, Marchand, and Pestieau 2000; Cremer, Pestieau, and Rochet 2003) may all contribute to the difference between the cross-sectional profile $s'(z)$ and the causal income effect $s'_{inc}(z)$. A key feature of our sufficient statistics approach is that the model can be reinterpreted so that this difference represents these alternative sources of heterogeneity—or a combination of them—and the characterization of optimal tax schedules remains intact.

To formalize this idea, consider the following modifications to our baseline model. First, assume that individuals might face type-dependent budget constraints given by $c \leq B(s,z,\chi;\theta) - \mathcal{T}(s,z)$, where χ represents a vector of auxiliary individual choices that may affect their budget. Second, assume that individuals may manipulate their taxable earnings z and taxable savings s or alter their ability to produce taxable earnings z by making productivity-enhancing investments. Consequently, individuals pay a tax of $\mathcal{T}(s,z)$, but their actual consumption of s and z is given by $\phi_s(s,z,\chi;\theta)$ and $\phi_z(s,z,\chi;\theta)$. Individuals' utility is thus given by $U(c,\phi_s(s,z,\chi;\theta),\phi_z(s,z,\chi;\theta),\chi;\theta)$.

We establish the following equivalence result.

[20] When there is important heterogeneity within income levels, the optimal mechanism may also involve deviations from smooth tax systems, for example, in the form of imposed bunching as in Moser and Olea de Souza e Silva (2019). Such cases are not covered by our multidimensional tax formulas: they only apply to smooth tax systems.

LEMMA 2: *For any given tax system* $\mathcal{T}(s,z)$, *individuals make identical choices in* (i) *an economy in which they have type-dependent preferences* $U(c,\phi_s(s,z,\chi;\theta),\phi_z(s,z,\chi;\theta),\chi;\theta)$ *and type-dependent budget domains* $B(s,z,\chi;\theta)$ *that may be affected by auxiliary choices* χ *and in* (ii) *an economy in which individuals have type-independent budget domains* $B(s,z)$, *no auxiliary choices, and type-dependent preferences*

$$(26) \quad \tilde{U}(c,s,z;\theta)$$

$$:= \max_{\chi} U(c + B(s,z,\chi;\theta) - B(s,z),\phi_s(s,z,\chi;\theta),\phi_z(s,z,\chi;\theta),\chi;\theta).$$

Lemma 2 shows that an economy that features preference heterogeneity and budget heterogeneity as well as auxiliary choices is equivalent to an economy with preference heterogeneity only, provided that preferences are suitably defined. The intuition is that under a given tax system $\mathcal{T}(s,z)$, individuals' utility maximization problems are equivalent in both economies. This means that all individuals make identical choices and attain the same level of utility, and the government collects the same tax revenue.[21] Since this equivalence holds for any tax system $\mathcal{T}(s,z)$, it immediately follows that these equivalent economies share the same optimal tax system, leading to the following proposition.

PROPOSITION 1: *In an economy with preference heterogeneity, budget heterogeneity, and auxiliary choices, an optimal smooth tax system remains characterized by Theorem* 1 *or Theorem* 2 *as long as the equivalent economy with only preference heterogeneity satisfies the assumptions of those theorems.*

Proposition 1 clarifies the generality of our sufficient statistics characterization of optimal taxes, where the sufficient statistic $s'(z) - s'_{inc}(z)$, or its multidimensional analog, captures all relevant dimensions of across-income heterogeneity that justify taxing s. While these different sources of across-income heterogeneity have previously been studied in isolation to qualitatively assess the robustness of the Atkinson-Stiglitz Theorem, our sufficient statistics techniques can be applied to account for them in a quantitative and general manner. To emphasize this point, online Appendix B.A provides a structural characterization of this sufficient statistic in economies with preference heterogeneity, budget heterogeneity, and auxiliary choices.

The intuition for this generality stems from the logic of Feldstein (1999), who shows that the elasticity of taxable income is a sufficient statistic for efficiency losses irrespective of whether it is due to real labor supply responses or avoidance

[21] This holds because the tax $\mathcal{T}(s,z)$ is measured in dollars and paid out of earnings. If instead the tax had a two-part structure where individuals pay T_1 in units of c (e.g., dollars) and T_2 in units of s (e.g., liters of soda), then budget heterogeneity would affect tax revenue. Such a system is relatively common for savings, where taxes are often paid in units of "period 2" dollars, after returns have been realized. This creates an additional arbitrage motive to tax individuals with higher returns more heavily in units of s. We explore such arbitrage motives in Section VIB.

behavior.[22] This logic spans a wide range of static and steady-state models with various forms of utility functions and budget constraints.[23] We now discuss some examples of budget heterogeneity and auxiliary choices, focusing on those that are relevant in the context of savings taxes.

Heterogeneous Prices.—One widespread form of budget heterogeneity is price heterogeneity. Suppose that individuals face prices $p(s,z,\chi;\theta)$ for s that depend on their level of s, their earnings z, their effort χ to seek out lower prices, or their type θ. Their budget domain is then $B(s,z,\chi;\theta) = z - p(s,z,\chi;\theta)s$, and their preferences are $U(c,s,z,\chi;\theta)$, allowing for monetary or psychic costs of effort χ. For instance, in the context of savings taxation, higher savings might allow individuals to lock in better interest rates, higher income might generate network effects that expose individuals to better opportunities (an example of "scale dependence"), and higher-ability types may be better at finding lower prices or higher returns on investments (an example of "type dependence").

Lemma 2 shows that this economy is equivalent to an economy with budget domain $B(s,z) := z - s$, with the price normalized to $p \equiv 1$, and where individuals' utility function is defined as $\tilde{U}(c,s,z;\theta) := \max_\chi U(c + [1 - p(s,z,\chi;\theta)]s, s, z, \chi;\theta)$. Intuitively, with a price of $p = p'$ instead of $p = 1$, individuals enjoy $(1 - p')s$ more consumption of c at a given s. This also shows that in our baseline model, the price p can be normalized to 1 without loss of generality—a feature we employ in some online Appendix proofs.

A key insight is that some sources of across-income price variation justify taxing s, while others do not, and the decomposition of the across-income profile $s'(z)$ into income effects, $s'_{inc}(z)$, and all other sources correctly distinguishes between them. In particular, *typ*e-dependent variation in prices will generally lead to $s'(z) \neq s'_{inc}(z)$ and thus to deviations from the Atkinson-Stiglitz result, whereas *scale*-dependent variation in prices would contribute to both $s'(z)$ and $s'_{inc}(z)$ and would thus preserve the Atkinson-Stiglitz result.

Heterogeneous Endowments.—Another form of budget heterogeneity is differences in initial endowments. Suppose that individuals have endowments $y_0(\theta)$, e.g., from inheritances or early-life capital returns, such that their budget domain is $B(s,z,\chi;\theta) = z + y_0(\theta) - s$. In the presence of income effects, endowments would influence individuals' decisions, and *typ*e-dependent endowments would therefore enter in the difference $s'(z) - s'_{inc}(z)$. For instance, this could represent a situation in which higher-productivity individuals might have more savings in part because they have larger inheritances, resulting in $s'(z) - s'_{inc}(z) > 0$ and justifying taxes on s.

[22] Chetty (2009) suggests limitations to Feldstein's (1999) results due to some avoidance behaviors generating new types of fiscal externalities or due to behavioral biases. Variations of these considerations are relevant in our setting as well, as explored in Sections VIA and VIB, respectively.

[23] These equivalent classes of models are static or steady state. Therefore, issues related to progressive information revelation and transition dynamics require a fuller dynamic model; while the strategy of quantifying heterogeneity using the discrepancy between cross-income differences and causal income effects may still be useful in such settings, we leave this possibility to future research.

Income Shifting.—Suppose that some auxiliary actions allow individuals to shift some of their labor income to capital income. Let $\tilde{z}$ and $\tilde{s}$ be the individuals' true labor income and savings, which are unobserved by the tax authority. Let χ denote the amount of labor income $\tilde{z}$ that individuals shift to be realized as taxable savings (including capital income), and let $m(s,z,\chi;\theta)$ denote any financial costs involved in income shifting. Individuals' taxable labor income is thus $z = \tilde{z} - \chi$, and their taxable savings are $s = \tilde{s} + \chi$. This describes an economy with budget domain $B(s,z,\chi;\theta) = z - s + 2\chi - m(s,z,\chi;\theta)$ and preferences $U(c, s - \chi, z + \chi, \chi;\theta)$, where χ might directly influence utility because of effort or psychic costs. In this example, $\phi_s(s,\chi) = s - \chi$ and $\phi_z(z,\chi) = z + \chi$. If individuals with higher earnings ability are better able to engage in income shifting, this type dependence will effectively translate in $s'(z) - s'_{inc}(z) > 0$. In contrast, any scale dependence in income shifting would enter both $s'(z)$ and $s'_{inc}(z)$.

Human Capital Investments.—Our framework can also be applied to issues of human capital investments, which is often modeled as affecting productivity later in life (Bovenberg and Jacobs 2005; Stantcheva 2017). As shown in Bovenberg and Jacobs (2005), some insights can be drawn from a static (steady-state) model in which the productivity-enhancing effect of human capital may be directly captured in preferences through, e.g., $\phi_z(s,z;\theta)$, where s represents human capital investments like education. This setup can be spanned by our model: using Lemma 2, such an economy is equivalent to one with preferences $\tilde{U}(c,s,z;\theta) := U(c,s,\phi_z(s,z;\theta);\theta)$.

B. *Measuring* s'_{inc}

The key sufficient statistic $s'(z) - s'_{inc}(z)$, or its multidimensional analog, can be obtained from familiar empirical strategies. The term $s'(z)$, and its multidimensional version, represents variation in s across the income distribution, which can be measured using standard data sources. The statistic s'_{inc} can be estimated using a variety of strategies. Here, we present three methods of measuring s'_{inc}, which rely on different types of quasi-experimental variation and are valid with both unidimensional or multidimensional heterogeneity.

PROPOSITION 2: *Define $\xi^s_w(\theta)$ as the elasticity of s with respect to the wage rate w, $\xi^h_w(\theta)$ as the elasticity of hours with respect to the wage rate, and $\varsigma^c_s(\theta)$ as the elasticity of s with respect to the marginal net-of-tax rate on labor income. The sufficient statistic $s'_{inc}(\theta)$ can be measured as follows:*

M1. If preferences are weakly separable and the tax system is separable, $s'_{inc}(\theta) = \dfrac{1 - T'_z(z(\theta))}{1 + T'_s(s(\theta))}\eta_{s|z}(\theta).$

M2. If wage rates w and hours h are observable, $s'_{inc}(\theta) = \dfrac{s(\theta)}{z(\theta)}\dfrac{\xi^s_w(\theta)}{1 + \xi^h_w(\theta)}.$

M3. If s-responses and z-responses to earnings tax reforms are estimable, $s'_{inc}(\theta) = \dfrac{s(\theta)}{z(\theta)}\dfrac{\varsigma^c_s(\theta)}{\varsigma^c_z(\theta)}.$

If individuals' preferences are weakly separable and if the tax system in place is separable in s and z, then s'_{inc} is proportional to the income effect parameter for s. If individuals' preferences are not weakly separable but wage rates w and hours h are observable, s'_{inc} can be related to the elasticity of s with respect to the wage rate and to the elasticity of hours with respect to the wage rate. If the elasticities of both s and z with respect to the marginal tax rate on z are observable, s'_{inc} can be recovered from these elasticities.

A key question for empirical implementation is the time horizon over which the statistics must be measured. Interpreting our static model to represent a steady-state economy, s'_{inc} corresponds to the causal effect of a change in steady-state labor income on steady-state consumption of s.[24] Therefore, the s'_{inc} parameter of interest is the *long-run* response of s to a change in labor income. If some determinants of s-consumption respond to income changes with a lag—for example, due to consumption commitments or because some preferences are endogenous to income but take time to adjust—then those delayed adjustments should be viewed as a component of s'_{inc} through the lens of this static model. Under the weak separability assumption, it is therefore necessary to measure the *long-run* marginal propensity to consume s. In the case of savings, this is the long-run marginal propensity to save, as estimated by Fagereng, Holm, and Natvik (2021), for example, in response to a change in unearned income.[25]

V. Optimal Simple Tax Systems

In practice, policymakers and institutions may impose constraints on the degree of complexity in the tax function. In this section, we apply our sufficient statistics methods to characterize the optimal policy for a few classes of tax systems with restrictions such as separability or linearity.

A. *A Taxonomy of Common Simple Savings Tax Systems*

Many governments tax both labor income and savings (or capital interest income). While these tax systems take various forms, the details of which depend on specifics such as timing, many can be interpreted as a function of earnings and savings, analogous to our function $\mathcal{T}(s,z)$. Table 1 presents three classes of simple tax systems: separable linear (SL), separable nonlinear (SN), as studied in Section I, and linear earnings-dependent (LED).

[24] A natural question is whether the effect of income received earlier in life (e.g., family income in childhood) should be used to measure the long-run income effect s'_{inc}. It should not. As shown by Lemma 1, the role of s'_{inc} is to quantify the distortion in work-life income induced by a change in the steady-state tax on s, and this distortion depends on the causal effect of earnings *during work life* on s. To the extent that income earlier in life affects s consumption differently from income during work life, the former behaves like a component of preference heterogeneity.

[25] Fagereng, Holm, and Natvik (2021) use lottery winnings as a source of exogenous variation in unearned income. However, if individuals respond differently to a one-time change in unearned income than to a persistent change of equal present value, then $s'_{inc}(z)$ should be measured based on the latter. We discuss this issue, and alternative measures of $s'_{inc}(z)$ based on survey data, in Section VII. There is some evidence that mental accounting and other behavioral frictions affect people's propensity to consume and save out of windfalls (e.g., Thakral and To 2021). Since steady-state changes in earnings correspond to anticipated changes in earnings, unanticipated windfalls could lead to an underestimate of s'_{inc} when s corresponds to savings and an overestimate when s corresponds to immediate consumption.

Table 1—Types of Simple Tax Systems

Type of simple tax system	$\mathcal{T}(s,z)$	$\mathcal{T}_s'(s,z)$	$\mathcal{T}_z'(s,z)$
SL: separable linear	$\tau_s s + T_z(z)$	τ_s	$T_z'(z)$
SN: separable nonlinear	$T_s(s) + T_z(z)$	$T_s'(s)$	$T_z'(z)$
LED: linear earnings-dependent	$\tau_s(z)s + T_z(z)$	$\tau_s(z)$	$T_z'(z) + \tau_s'(z)s$

Online Appendix Table A1 categorizes tax policies on each of 5 classes of savings-related tax bases—wealth, capital gains, property taxes, pensions, and inheritances—for 21 countries, showing that most of these taxes can be understood as fitting into 1 of the 3 simple tax system classes from Table 1. In cases where there is ambiguity, we provide supplementary information.[26]

In the United States, for example, most property taxes, levied at the state and local level, take the form of a separable linear tax: a flat tax rate, independent of one's labor earnings, is applied to the assessed value of the total property. The estate tax takes the form of a separable nonlinear tax: tax rates rise progressively with the value of the estate, but they do not vary with labor income of the donor or the recipient. Taxes on long-term capital gains and qualified dividends take the form of linear earnings-dependent taxes.[27] In 2020, for example, an individual with $50,000 in labor earnings faced a linear tax of 15 percent on long-term capital gains, whereas an individual earning $500,000 faced a linear tax of 20 percent. Finally, although we focus on savings tax policies, these classes of simple tax systems are also relevant for other classes of commodities. Separable linear commodity taxes are ubiquitous (e.g., on lodging, airfare, and sales taxes that apply to specific classes of consumption), while separable nonlinear and linear income-dependent tax structures are often used for taxes (or subsidies) on goods like energy and education.[28]

B. *Optimal Simple Tax Systems under Unidimensional Heterogeneity*

We now present optimality conditions for SL, SN, and LED tax systems under the assumption that heterogeneity is unidimensional. The optimality condition for the nonlinear tax on earnings, T_z, is the same as condition (18) presented in Theorem 2.

[26] We impose several simplifications for our interpretation of the tax codes. First, we treat ordinary income as consisting primarily of labor income (earnings), written as z in our notation. Second, we separately consider taxes on five broad categories of savings vehicles: wealth, capital gains, real property, private pensions, and inheritances. These categories may overlap (e.g., real property is a component of wealth) but we use these groups to reflect the tax instruments that governments use in practice.

[27] This is a slight approximation since the linear capital gains tax rate in the United States is a function of total income (including capital gains) rather than solely labor income.

[28] One practical distinction between taxes on savings and on other commodities involves the measurement of the tax base. In our baseline model, the argument of the tax function s represents the quantity of s consumed. This is natural for many commodities, but in the setting of savings, it is common for the tax system to be written as a function of gross savings before taxes, e.g., a tax $T_1(z)$ in period 1 and a tax $T_2(s_g,z)$ on gross pretax savings $s_g = (1+r)(z - T_1(z) - c)$ in period 2, where r is the compounded rate of return, so that period 2 consumption is given by $s = s_g - T_s(s_g,z)$. In this formulation, an SL structure is one where $T_2(s_g,z) = \tau_s s_g$, an SN structure is one where $T_2(s_g,z)$ is a function of s_g only, and an LED structure is one where $T_2(s_g,z) = \tau_s(z)s_g$. Fortunately, there is an equivalence between these formulations of two-period tax systems and the type of "static" tax function $\mathcal{T}$ considered in our baseline model, as we show in online Appendix B.E.

We thus focus on optimal marginal tax rates on s, for which it is convenient to use the "bar" notation, as in $\overline{\zeta^c_{s|z}}$, to denote a population average.

PROPOSITION 3: *Consider an optimal SL/SN/LED tax system for which Conditions 1 and 2-UD hold. Suppose also that in the SN system s is strictly monotonic in z. At each bundle* $\left(c^0, s^0, z^0\right)$ *chosen by a type* θ, *this tax system satisfies*

$$SL: \frac{\tau_s}{1+\tau_s} = \frac{1}{\overline{s}\,\overline{\zeta^c_{s|z}}} \int_z \left[s'(z) - s'_{inc}(z)\right]\left[\int_{x \geq z}\left[1-\hat{g}(x)\right] dH_z(x)\right] dz \tag{27}$$

$$= -\frac{1}{\overline{s}\,\overline{\zeta^c_{s|z}}} \operatorname{cov}\left[\int_{x \leq z}\left[s'(x) - s'_{inc}(x)\right] dx, \hat{g}(z)\right] \tag{28}$$

$$SN: \frac{T'_s\left(s^0\right)}{1+T'_s\left(s^0\right)} = \left[s'\left(z^0\right) - s'_{inc}\left(z^0\right)\right] \frac{1}{s^0 \zeta^c_{s|z}\left(s^0\right)} \frac{1}{s'\left(z^0\right) h_s\left(s^0\right)} \times \int_{s \geq s^0}\left[1-\hat{g}(s)\right] dH_s(s) \tag{29}$$

$$LED: \frac{\tau_s\left(z^0\right)}{1+\tau_s\left(z^0\right)} = \left[s'\left(z^0\right) - s'_{inc}\left(z^0\right)\right] \frac{1}{s^0 \zeta^c_{s|z}\left(z^0\right)} \frac{1}{h_z\left(z^0\right)} \times \int_{z \geq z^0}\left[1-\hat{g}(z)\right] dH_z(z). \tag{30}$$

Consider an SL/SN/LED system that is not Pareto dominated by another SL/SN/LED system and for which Conditions 1 and 2-UD hold. Suppose also that in the SN system s is strictly monotonic in z. At each bundle $\left(c^0, s^0, z^0\right)$ *chosen by a type* θ, *this tax system satisfies*

$$SL: \frac{\tau_s}{1+\tau_s} = \frac{1}{\overline{s}\,\overline{\zeta^c_{s|z}}} \int_z \left[s'(z) - s'_{inc}(z)\right] z \zeta^c_z(z) \times \frac{T'_z(z) + s'_{inc}(z)\tau_s}{1-T'_z(z)} dH_z(z) \tag{31}$$

$$SN: \frac{T'_s\left(s^0\right)}{1+T'_s\left(s^0\right)} = \left[s'\left(z^0\right) - s'_{inc}\left(z^0\right)\right] \times \frac{z^0 \zeta^c_z\left(z^0\right)}{s^0 \zeta^c_{s|z}\left(s^0\right)} \frac{T'_z\left(z^0\right) + s'_{inc}\left(z^0\right) T'_s\left(s^0\right)}{1-T'_z\left(z^0\right)} \tag{32}$$

$$LED: \frac{\tau_s\left(z^0\right)}{1+\tau_s\left(z^0\right)} = \left[s'\left(z^0\right) - s'_{inc}\left(z^0\right)\right] \times \frac{z^0 \zeta^c_z\left(z^0\right)}{s^0 \zeta^c_{s|z}\left(z^0\right)} \frac{T'_z\left(z^0\right) + \tau'_s\left(z^0\right) s^0 + s'_{inc}\left(z^0\right)\tau_s\left(z^0\right)}{1-T'_z\left(z^0\right) - \tau'_s\left(z^0\right) s^0}. \tag{33}$$

The optimal tax formulas and the Pareto efficiency conditions for SN and LED systems are analogous to the conditions for $\mathcal{T}_s'$ in the fully flexible smooth tax systems derived in Theorem 1.[29] Intuitively, this is because these systems allow for different marginal tax rates on s across individuals. Additionally, online Appendix D.B presents sufficient conditions for SN and LED systems to implement the optimal allocation. In the case of SN systems, these conditions are relatively weak, although they do require that s is strictly monotonic with z in the optimal allocation. In contrast, the sufficient conditions for LED systems don't require monotonicity of s but, loosely speaking, require that the local preference for s must not vary "too strongly" across incomes.

The SL system is the most restrictive and generically cannot implement the same allocation as the optimal smooth tax system. This is because the optimal smooth tax system does not generally feature constant marginal tax rates on s. Correspondingly, the optimal tax formula for the SL systems takes a different form. As shown in equation (27), the constant marginal tax rate τ_s for SL systems is in a certain sense an average of the z-earner specific marginal tax rates in equation (30). Intuitively, the constant marginal tax rate is a population aggregate of the tax rates that would be optimal for individuals with different earnings levels. This is mirrored in the Pareto efficiency condition (31). Equation (28) expresses this optimality condition in an alternative way, which was first derived by Allcott, Lockwood, and Taubinsky (2019). This formulation has a familiar form resembling the Diamond (1975) "many-person Ramsey tax rule." It is identical to the Diamond (1975) formula when $\int_{x \leq z} \left[s'(x) - s_{inc}'(x) \right] dx = s(z)$; that is, when all consumption differences justify taxes on s.[30] In contrast, when all consumption differences are driven by heterogeneity in income such that $\int_{x \leq z} \left[s'(x) - s_{inc}'(x) \right] dx \equiv 0$, it reduces to the original statement of the Atkinson-Stiglitz Theorem. More generally, even for arbitrarily nonlinear taxes on s, the optimal tax rate is inversely proportional to the elasticity $\zeta_{s|z}^{c}$, consistent with standard Ramsey principles.

C. *Optimal Simple Tax Systems under Multidimensional Heterogeneity*

We next extend our characterization of optimal simple tax systems to settings with multidimensional heterogeneity. As with the unidimensional case, the optimality condition for the nonlinear tax on earnings, T_z, is analogous to the condition (24) presented in Theorem 2, except that expectations are taken over all individuals with the same earnings.[31] We thus focus on optimal marginal tax rates on s.

[29] We express (29) using an integral in the s-space rather than the z-space to make the formula more intuitive, but this is formally equivalent given the assumed existence of a mapping between s and z for SN tax systems under unidimensional heterogeneity. See Gerritsen et al. (2020) and Scheuer and Slemrod (2021) for a variation of the Pareto efficiency condition for SN systems for the case of heterogeneity in initial wealth and in rates of returns, respectively.

[30] In this special case, the SN optimal tax formula (29) also nests the optimal nonlinear tax formula of Saez and Stantcheva (2018), where wealth choices originate entirely from preferences for wealth. Use $h_s(s(z)) = \frac{h_z(z)}{s'(z)}$ to recover the equivalence with formula (11) in Saez and Stantcheva (2018).

[31] The formula for optimal marginal tax rates on z is $\mathbb{E}\left[FE_z | z^0\right] = \frac{1}{h_z(z^0)} \int_{z \geq z^0} \mathbb{E}\left[1 - \hat{g} | z\right] dH_z(z)$; see online Appendix B.C.

PROPOSITION 4: *Consider an optimal SL/SN/LED tax system for which Conditions 1 and 2-UD hold. At each bundle* (c^0, s^0, z^0) *chosen by a type* θ*, this tax system satisfies*

$$SL: \frac{\tau_s}{1+\tau_s} = \frac{1}{\overline{s}\,\overline{\zeta^c_{s|z}}}\Big\{ \int_z \Big[\Big(\overline{s}'(z) - \overline{s'_{inc}}(z) \Big) \int_{x \geq z} \mathbb{E}[1 - \hat{g} \,|\, x]\, dH_z(y) \Big] dz - \int_z \Big(\mathrm{cov}\big[\hat{g}, s|z\big] + \mathrm{cov}\big[FE_z, s'_{inc}|z\big] \Big) dH_z(z) \Big\}, \tag{34}$$

$$SN: \frac{T'_s(s^0)}{1+T'_s(s^0)} = \frac{1}{\mathbb{E}\big[s\,\zeta^c_{s|z}|s^0\big]}\Big\{ \frac{1}{h_s(s^0)} \int_{s \geq s^0} \mathbb{E}\big[1 - \hat{g}|s\big] dH_s(s) - \overline{s'_{inc}}(s^0)\mathbb{E}\big[FE_z \,|\, s^0\big] - \mathrm{cov}\big[FE_z, s'_{inc} \,|\, s^0\big] \Big\}, \tag{35}$$

$$LED: \frac{\tau_s(z^0)}{1+\tau_s(z^0)} = \frac{1}{\mathbb{E}\big[s\,\zeta^c_{s|z}\,|\,z^0\big]}\Big\{ \Big[\overline{s}'(z^0) - \overline{s'_{inc}}(z^0)\Big] \frac{1}{h_z(z^0)} \times \int_{z \geq z^0} \mathbb{E}[1 - \hat{g} \,|\, z] dH_z(z) - \mathrm{cov}\big[\hat{g}, s \,|\, z^0\big] - \mathrm{cov}\big[FE_z, s'_{inc} \,|\, z^0\big] + \frac{1}{h_z(z^0)} \frac{d}{dz^0}\Big[\mathrm{cov}\big[FE_z, s \,|\, z^0\big]\, h_z(z^0)\Big] \Big\}. \tag{36}$$

Formula (36) for optimal tax rates on s in an LED system resembles formula (25) presented in Theorem 2. This is because in an LED system both tax rates on z and tax rates on s are conditional on individuals' earnings z, allowing for joint tax reforms affecting the same individuals. The only difference is that tax rates are now constrained to be the same for all individuals with a given earnings level; thus, the term $\frac{d}{dz^0}\big(\mathbb{E}\big[s \,\big|\, s \geq s^0, z^0\big]\big) - \mathbb{E}\big[s'_{inc} \,|\, s \geq s^0, z^0\big]$ from Theorem 2 reduces to $\overline{s}'(z^0) - \overline{s'_{inc}}(z^0)$.

Formula (34) for the optimal linear tax rate on s in an SL system can once again be seen as an average of the z-earner-specific marginal tax rate on s from the LED formula. It features only two covariance terms because the third covariance term capturing the distortions associated with the phasing in of LED reforms around a given earnings level z^0 drops out.

The optimality condition (35) for the optimal nonlinear tax rate on s in an SN system cannot be written in a manner that resembles the conditions from Theorem 2. This is because in an SN system, increasing the marginal tax rates on s at s^0 versus on z at z^0 affects different sets of individuals; thus, we cannot combine reforms of tax rates on s and on z to produce simpler formulas. Condition (35) reduces to the simpler condition (29) only with unidimensional heterogeneity, in which case the covariance term vanishes and, by rearranging the expression for optimal income tax rates, $\mathbb{E}[FE_z \,|\, s^0] = \frac{1}{s'(z^0)\, h_s(s^0)} \int_{s \geq s^0} [1 - \hat{g}(s)]\, dH_s(s)$.

Finally, while we characterized optimal taxes on s under the assumption of an optimal earnings tax, the domain of simple tax systems allows us to also consider cases where the tax on z is not necessarily optimal. Online Appendix B.B provides formulas for optimal tax rates on s in SL, SN, and LED systems for a given, potentially suboptimal, earnings tax.

VI. Extensions

In this section we provide two extensions that are particularly relevant to the tax treatment of savings. First, we allow for the possibility that the government wants people to save more than their perceived private optima, either because of a misalignment between private and social intergenerational preferences or because of individuals' behavioral biases. Second, we consider the case where taxes can be collected both in units of c and in units of s, as is often the case for savings taxes. Additionally, online Appendix B.D generalizes our results to more than two dimensions of consumptions. This allows us to cover settings where, for example, individuals have access to multiple saving vehicles that are taxed differentially.

A. *Optimal Taxation with Corrective Motives*

Our framework can be interpreted as a bequest model in which parents work and consume in the first period and leave a bequest to their heirs in the second period. Under this interpretation, our baseline model makes the implicit assumption that the government values bequests in the same way as parents. Farhi and Werning (2010) consider a model where the weight that parents attach to the well-being of future generations is too low from a normative perspective. This misalignment introduces a motive to encourage bequests, which we consider in this extension.

Following Farhi and Werning (2010), we assume additively separable preferences given by

$$U(c,s,z;\theta) = u(c;\theta) - k(z;\theta) + \beta v(s;\theta), \tag{37}$$

where $u(c;\theta)$ is the utility parents derive from consumption c, $k(z;\theta)$ is the disutility parents incur to obtain earnings z, $v(s;\theta)$ is the utility heirs derive from a bequest s, and β is the weight parents attach to the well-being of their heirs. As in Farhi and Werning (2010), the government maximizes

$$\int_\theta \Big[U\big(c(\theta),s(\theta),z(\theta);\theta\big) + \nu v\big(s(\theta);\theta\big)\Big] dF(\theta), \tag{38}$$

where ν parameterizes the degree of misalignment. Farhi and Werning (2010) microfound ν as the Lagrange multiplier associated with a constraint that the future generation attains a required level of well-being $\int_\theta v\big(s(\theta);\theta\big)\, dF(\theta) \geq \underline{V}$.

This model can also be interpreted more generally to analyze behavioral biases or externalities as a motivation for encouraging savings. For example, suppose that $v(s;\theta) = \delta(\theta)u(s;\theta)$, where δ is the "exponential discount factor" and β is "present focus," as in Laibson (1997). If the government utilizes the "long-run criterion" for welfare, then the degree of misalignment is given by $\nu = (1-\beta)$.[32] More generally, β may be heterogeneous, so that misalignment is type dependent and given by $\nu(\theta) = (1-\beta(\theta))$.[33] Alternatively, there could be "macro" externalities if, for example, the higher returns to investing achievable by some individuals produce a net gain for society (Howitt and Aghion 1998).

Below, we characterize optimal taxation with heterogeneous misalignment, where $\beta(z)$ and $\nu(z)$ denote the parameters corresponding to a z-earner. This generalizes the result in Farhi and Werning (2010) by allowing heterogeneity in preferences for s and heterogeneity in the misalignment parameter ν.

PROPOSITION 5: *Consider an optimal tax system for which Conditions 1 and 2-UD hold. At each bundle (c^0, s^0, z^0) chosen by a type θ, this tax system satisfies*

$$\frac{\mathcal{T}_s'(s^0,z^0)}{1+\mathcal{T}_s'(s^0,z^0)} = \left[s'(z^0) - s_{inc}'(z^0)\right]\frac{1}{s^0\,\zeta_{s|z}^c(z^0)}\frac{1}{h_z(z^0)} \times \int_{z\geq z^0}\left[1-\hat{g}(z)\right]dH_z(z) - \frac{\nu(z^0)}{\beta(z^0)}g(z^0). \tag{39}$$

Consider a Pareto-efficient tax system for which Conditions 1 and 2-UD hold. At each bundle (c^0, s^0, z^0) chosen by a type θ, this tax system satisfies

$$\frac{\mathcal{T}_s'(s^0,z^0)}{1+\mathcal{T}_s'(s^0,z^0)} = \left[s'(z^0) - s_{inc}'(z^0)\right]\frac{z^0\,\zeta_z^c(z^0)}{s^0\,\zeta_{s|z}^c(z^0)} \times \left[\frac{\mathcal{T}_z'(s^0,z^0) + s_{inc}'(z^0)\,\mathcal{T}_s'(s^0,z^0)}{1-\mathcal{T}_z'(s^0,z^0)} + s_{inc}'(z^0)\frac{\nu(z^0)}{\beta(z^0)}g(z^0)\right] - \frac{\nu(z^0)}{\beta(z^0)}g(z^0). \tag{40}$$

This is an extension of Theorem 1, where the key new term is a form of Pigovian correction given by $\frac{\nu(z)}{\beta(z)}g(z)$. As equation (39) shows, the presence of misalignment motivates the government to lower the tax rate on s. The degree by which the government lowers the tax rate depends on the degree of misalignment (relative to the discount factor β) and on the social marginal welfare weight. Because welfare weights decline with z, equation (39) gives the "progressive estate taxation" result of Farhi and Werning (2010)—i.e., savings subsidies that decline with income—under

[32] See Bernheim and Taubinsky (2018) for a detailed discussion of such a criterion, as well as alternative normative approaches to studying the implications of present focus.

[33] For example, Lockwood (2020) summarizes evidence suggesting that individuals with higher earnings ability have lower degrees of present focus.

the special assumptions that (i) $s'(z) = s'_{inc}(z)$ and (ii) $\beta(z) \equiv \beta \in \mathbb{R}$, $\nu(z) \equiv \nu \in \mathbb{R}$. This core result of Farhi and Werning (2010) extends the standard Pigovian taxation logic to optimal screening of distortions with a nonlinear tax.

More generally, Proposition 5 provides a simple formula for balancing the "corrective motives" studied by Farhi and Werning (2010) with the additional motives to tax s in the presence of preference heterogeneity studied in this paper. This extends the Allcott, Lockwood, and Taubinsky (2019) results for linear commodity taxes with biased consumers to study optimal screening of biases with a nonlinear tax. If $s'(z) > s'_{inc}(z)$ and $\nu(z)/\beta(z)$ and $g(z)$ are decreasing with z, Proposition 5 suggests a progressive tax on s that can feature subsidies at low incomes and taxes at high incomes.

B. *Tax Arbitrage with Heterogeneous Prices*

So far, we have considered tax functions where the tax is always paid in units of the numeraire good c. In some applications it is also natural to consider tax systems where taxes are also collected in units of s. When p is heterogeneous and tax systems are not fully flexible, this produces a "tax arbitrage" motive pushing for relatively higher savings taxes (and lower earnings taxes) on individuals with higher rates of returns. This extension provides a generalization of our baseline results and relates to independent work by Gerritsen et al. (2020).

Formally, suppose that the government uses a two-part tax structure, where individuals pay a tax $T_1(z)$ in units of c (e.g., earnings tax levied in period 1) and a tax $T_2(s,z)$ in units of s (e.g., savings tax levied in period 2). For concreteness, we refer to T_1 as period 1 taxes and to T_2 as period 2 taxes. Following Gahvari and Micheletto (2016), we consider heterogeneous rates of transformation between c and s that are a function of gross earnings and type, $p(z,\theta)$. For example, wealthier individuals may have access to better rates of return on savings or prices of commodities. Alternatively, higher earnings ability may be associated with a better ability to obtain high rates of return or to find better prices.

Individuals maximize utility subject to the budget constraint $c + p(z,\theta)s \leq z - T_1(z) - p(z,\theta)\,T_2(s,z)$. We assume that Assumption 2-UD holds and denote by $\vartheta(z)$ the type θ of individuals who choose earnings z. We slightly abuse notation to define $p(z) := p(z,\vartheta(z))$. The government, as before, maximizes a weighted average of utilities subject to the constraints

$$\int_z T_1(z)\,dH_z(z) \geq E_1 \quad \text{and} \quad \int_z T_2(s(z),z)\,dH_z(z) \geq E_2, \tag{41}$$

which generate marginal values of public funds λ_1 and λ_2. We continue using $\hat{g}(z)$ to denote the social marginal welfare effect of increasing a z-earner's consumption of c by one unit, normalized by the marginal value of public funds λ_1.[34]

Heterogeneity in p generates efficiency effects through two channels. First, for individuals with relatively low $p(z)$, it is efficient for the government to decrease T_1 and increase T_2. Second, lump-sum changes in T_2 trigger novel substitution effects.

[34] The formula for $\hat{g}(z)$ in this more general setting is in online Appendix C.J.3.

This is because a lump-sum increase dT in T_2 has the same effect on an individual's utility as a $p(z)\,dT$ increase in T_1 and thus changes behavior as much as a marginal tax rate change of $\frac{\partial p}{\partial z} dT$ in T_1. We denote by

$$\varphi(z) := -\left(T_1'(z) + \frac{\lambda_2}{\lambda_1}\frac{\partial T_2}{\partial z} + s_{inc}'(z)\frac{\lambda_2}{\lambda_1}\frac{\partial T_2}{\partial s}\right)\frac{z\zeta_z^c(z)}{1 - T_1'(z)}\frac{\partial p}{\partial z}$$

the fiscal impacts of this substitution effect at earnings z. The impact of a uniform lump-sum change in T_2 is then $\overline{\hat{g}p - \varphi}$, where the "bar" notation is used to denote a population average across all earnings levels. Thus, $\lambda_2/\lambda_1 = \overline{\hat{g}p - \varphi}$, as we formally show in online Appendix C.J.3.

PROPOSITION 6: *Consider an optimal two-part SN/LED tax system for which Conditions 1 and 2-UD hold. Suppose also that in the SN system s is strictly monotonic in z. At each bundle (c^0, s^0, z^0) chosen by a type θ, this SN system satisfies*

$$\frac{\lambda_2/\lambda_1\, T_2'(s^0)}{1 + p(z^0)\,T_2'(s^0)} \tag{42}$$
$$= \frac{1}{s^0\,\zeta_{s|z}^c(z^0)}\,\frac{1}{h_z(z^0)}\Bigg\{\left[s'(z^0) - s_{inc}'(z^0)\right] \times \int_{z\geq z^0}\left[1 - \hat{g}(z)\right] dH_z(z)$$
$$+ \frac{s'(z^0)}{p(z^0)}\left[\Psi(z^0) + \int_{z\geq z^0}\left(\varphi(z) - \overline{\varphi}\right) dH_z(z)\right]\Bigg\},$$

where

$$\Psi(z^0) := \left[1 - H_z(z^0)\right]\int_{z\leq z^0}\hat{g}(z)\left[p(z) - p(z^0)\right] dH_z(z) \tag{43}$$
$$+ H_z(z^0)\int_{z\geq z^0}\hat{g}(z)\left[p(z^0) - p(z)\right] dH_z(z).$$

At each bundle (c^0, s^0, z^0) chosen by a type θ, this LED system satisfies

$$\frac{\lambda_2/\lambda_1\,\tau_s(z^0)}{1 + p(z^0)\,\tau_s(z^0)} \tag{44}$$
$$= \frac{1}{s(z^0)\,\zeta_{s|z}^c(z^0)}\,\frac{1}{h_z(z^0)}\left[s'(z^0) - s_{inc}'(z^0)\right] \times \int_{z\geq z^0}\left[1 - \hat{g}(z)\right] dH_z(z)$$
$$+ \frac{1}{\zeta_{s|z}^c(z^0)}\,\frac{1}{p(z^0)}\left\{\frac{p'(z^0)}{h_z(z^0)}\int_{z\geq z^0}\left[1 - \hat{g}(z)\right] dH_z(z) + \left[\overline{\hat{g}p} - \overline{\hat{g}}\,p(z^0) + \varphi(z^0) - \overline{\varphi}\right]\right\}.$$

Proposition 6 shows that the sufficient statistic $s'(z) - s'_{inc}(z)$ continues to play a central role for optimal taxes on s.[35] On the left-hand side of (42) and (44), the presence of $p(z)$ in the denominator is because an individual's marginal tax rate on s, translated to units of c, is $p(z)\frac{\partial T_2}{\partial s}$. The presence of λ_2/λ_1 in the numerator of the left-hand side is because fiscal externalities generated by substitution away from s must be weighted by the "period 2" marginal value of public funds.

Proposition 6 also introduces novel efficiency terms that lead to taxes on s, even when $s'(z) = s'_{inc}(z)$. In the SN formula, there are two additional efficiency effects. These terms are both positive and thus push toward taxing s when higher earners (i) face lower prices p (e.g., higher rates of returns on savings) and choose higher levels of s and (ii) exhibit larger substitution effects φ. The first term, proportional to $\Psi(z)$, captures the efficiency effects of increasing period 2 taxes. This term is unambiguously positive when p decreases cross-sectionally with z and captures the intuition that with an SN system, increasing marginal tax rates on s for higher earners increases period 2 taxes on individuals with below-average p. The second term, proportional to $\int_{x \geq z}\left[\varphi(x) - \overline{\varphi}\right] dH_z(x)$, captures the fact that increasing marginal tax rates on s motivates individuals to increase labor supply in order to get lower prices p when $\partial p/\partial z < 0$.[36]

The implications for LED tax systems are somewhat different. Assume again that p declines cross-sectionally with z (i.e., $p'(z) < 0$). The first novel term in equation (44), proportional to $p'(z)$, reflects the fact that higher earners are less responsive to marginal changes in T_2 when $p(z)$ declines with income since period 2 consumption is "cheaper" for them than period 1 consumption. The second term, proportional to $\overline{\hat{g}p} - \overline{\hat{g}}p(z) + \varphi(z) - \overline{\varphi}$, is also negative for sufficiently low values of z, as in this case both $\overline{\hat{g}p} - \overline{\hat{g}}p(z)$ and $\varphi(z) - \overline{\varphi}$ are negative. However, this term is positive for sufficiently high values of z. Thus, when $s'(z) = s'_{inc}(z)$, the optimal LED system features subsidies on s for lower-income individuals and taxes on s for higher-income individuals.

The contrast in implications for SN versus LED tax systems—everywhere-positive tax rates in the former, subsidies followed by taxes in the latter—highlights that the new efficiency considerations from heterogeneous rates of return depend on the types of restrictions imposed on the tax system. The reason for this dependence is because positive tax rates on s are a consequence of a missing instrument problem. In a fully flexible tax system, the efficiency gains of taxing a person in period 2 instead of period 1 could be obtained by shifting each individual's total tax burden onto their lowest-cost tax base up to the point that heterogeneous prices are arbitraged away, without the distortion of increasing marginal tax rates on s. But less flexible tax systems can only generate this shifting of the tax burden by altering marginal tax rates on s, and the optimal means of doing this depend on the nature of the restricted tax system.

[35] A characterization of the optimal earnings tax $T_1(z)$ is in online Appendix C.J.4.

[36] The SN formula generalizes the result in Gerritsen et al. (2020) to incorporate other forms of across-income heterogeneity and makes transparent the sign of these terms in a formula employing measurable sufficient statistics.

VII. Empirical Application

We apply our formulas to study the tax treatment of savings in the United States. We compute SL, SN, and LED schedules using calibrated models of the US economy, first assuming unidimensional heterogeneity and then allowing multidimensional heterogeneity. Section VIIA presents our baseline calibration and statistics that have already been studied extensively in prior literature, such as the elasticity of labor supply. Section VIIB focuses on the calibration of the key statistic s'_{inc}, which has received less focus in the empirical literature but which is central to quantifying across-income heterogeneity. Section VIIC presents the results of our simulations. As is typical for calculations based on sufficient statistics formulas, these results are approximations, as they do not account for changes in the underlying distributions and sufficient statistics that might arise if the savings tax were reformed.

A. *Calibration*

We calibrate a model of the US economy that can be interpreted through the lens of our model with a joint savings and income tax function $\mathcal{T}(s,z)$, expressed in terms of the three simple tax systems described in Table 1. Online Appendix E discusses details of this calibration; here, we summarize the key steps. We calibrate a two-period model economy with a fine grid of incomes, where the first period corresponds to work life and the second to retirement. We assume a constant (and, in our baseline, homogeneous) annual rate of return of 3.8 percent before taxes, drawing from Fagereng et al. (2020). We calibrate a version of the economy with unidimensional heterogeneity (i.e., a single level of savings at each income) and a version with multidimensional heterogeneity, reporting results for each below.

Because our model builds on standard models of commodity taxation, it implicitly assumes that z and $\mathcal{T}(s,z)$ are measured in the same units as consumption, which in a dynamic setting corresponds to "period 1" dollars. In practice, savings taxes are typically levied after returns, and they are thus measured in "period 2" dollars. We accordingly translate all tax rates into units of period 2 dollars when reporting results, so that a marginal savings tax rate of 10 percent indicates that if an individual's total wealth at retirement increases by \$1, then they must pay an additional \$0.10 in taxes when they retire.

To calibrate the earnings and savings distributions—and thus the across-income savings profile $s(z)$—we use the Distributional National Accounts micro-files of Piketty, Saez, and Zucman (2018). We use 2019 measures of pretax labor income (*plinc*) and net personal wealth (*hweal*) at the individual level, as well as the age category (20 to 44 years old, 45 to 64, and above 65). Discretizing the income distribution into percentiles by age group, our measure of annualized earnings during the working life z at the nth percentile is constructed by averaging earnings at the nth percentile across those aged 20 to 44 and those aged 45 to 64. Our measure of savings is calibrated using the profile of average net personal wealth, *hweal*, in each income percentile in the 45–64 age bucket, plotted in online Appendix Figure A1. This measure of wealth includes housing assets, business assets, and financial assets,

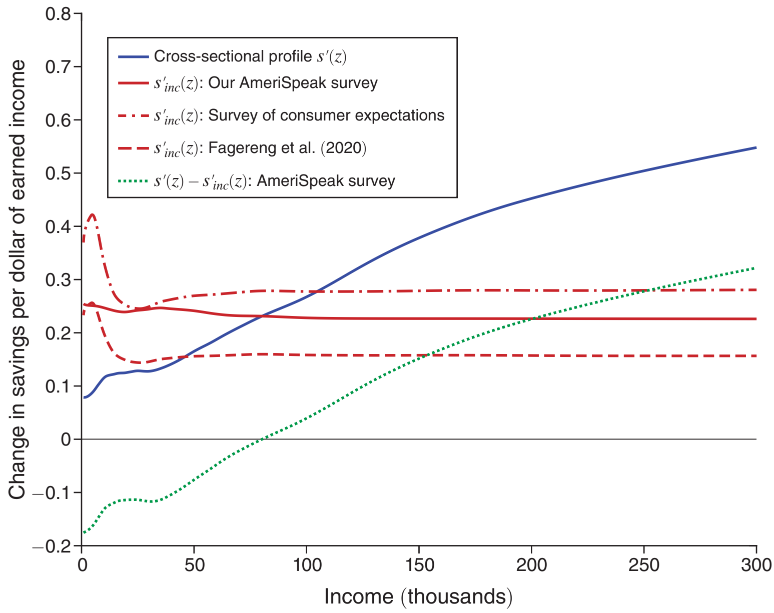

Figure 2. Decomposition of Cross-Sectional Savings Profile

Notes: This figure reports the slope of the cross-sectional profile of savings $s'(z)$ (blue) as well as our calibrations of $s'_{inc}(z)$ based on causal income effects, derived from Fagereng, Holm, and Natvik (2021) and from a new nationally representative survey. See Section VII and online Appendix E.A for details.

net of liabilities, as well as defined-contribution pension and life insurance assets.[37] It does not include social security, which we model as lump-sum forced savings that are received during retirement. The figure shows that the stock of savings upon retirement is approximately zero at low incomes but increases substantially with income.

We project this wealth profile forward to age 65 based on the growth rate observed in the Survey of Consumer Expectations (described below). We then convert to net-of-tax savings using a calibration of savings tax rates across the earnings distribution in the United States, derived by computing the weighted average of savings tax rates using the asset composition of savings portfolios reported in Bricker, Moore, and Thompson (2019); see online Appendix E.A.2 for details. Finally, we normalize both labor earnings and retirement savings by the number of years worked. This provides us with a population of representative individuals at each percentile of the income distribution, for whom period 1 represents their working life, with a representative age of 45, and period 2 represents retirement, which occurs 20 years later at age 65. The convex shape of the savings profile, which persists after accounting for taxes, indicates that the cross-sectional slope $s'(z)$ thus rises with income, as shown by the solid blue line in Figure 2.

[37] The ongoing methodological discussion regarding the different ways to measure wealth (see, e.g., Saez and Zucman 2020; Smith, Zidar, and Zwick 2021) has important implications for estimates of wealth in the top 1 percent but has little impact on the wealth distribution of the rest of the population that we are using here.

We assume a constant compensated earnings elasticity of $\zeta_z^c = 0.33$, drawn from the meta-analysis of Chetty (2012). The value of the savings elasticity $\zeta_{s|z}^c$ is related to the elasticity of taxable wealth (e.g., Jakobsen et al. 2020) and to the elasticity of capital gains realizations with respect to the capital gains tax (e.g., Agersnap and Zidar 2021). However, studies that measure elasticities from responses to tax reforms are likely inflated by cross-base responses, as taxpayers reoptimize their savings portfolio toward savings vehicles that are relatively less taxed after the reform.[38] We report results for a broad range of values spanning $\zeta_{s|z}^c = 0.7$ to $\zeta_{s|z}^c = 3$, with a baseline of $\zeta_{s|z}^c = 1$, which approximately aligns with the baseline calibration considered in Golosov et al. (2013), in which the intertemporal elasticity of substitution is set to 1. Online Appendix E.A.4 discusses the correspondence.

B. *Estimation of Across-Income Heterogeneity*

Having calibrated the across-income savings profile, $s'(z)$, our measure of local across-income heterogeneity requires the causal income effect s_{inc}', which we estimate using two complementary strategies.

The first estimation strategy is motivated by the Proposition 2 result that when preferences are weakly separable, the causal income effect of windfall income on s consumption identifies s_{inc}'. To the extent that separability is plausible, we can exploit exogenous shocks to unearned income in order to estimate s_{inc}'. To implement this strategy, we draw from Fagereng, Holm, and Natvik (2021), who estimate the marginal propensity to consume (MPC) out of windfall income across the earnings distribution using information on lottery prizes linked with administrative data in Norway.[39] Lottery consumption is widespread in Norway—over 70 percent of adults from all income groups participated in 2012—and administrative records of asset and wealth holdings allow for direct measures of savings and consumption responses to lottery winnings. Fagereng, Holm, and Natvik (2021) find that individuals' consumption peaks during the winning year and then gradually reverts to the pre-win baseline. Over a 5-year horizon, they estimate that winners consume close to 90 percent of the prize (see their Figure 2, "aggregate consumption response"), which translates into a long-run MPC of 0.9 and a marginal propensity to save of 0.1. They do not find significant heterogeneity across incomes in this MPC. We convert this homogeneous MPC into a response of net retirement savings to changes in pretax labor income using our calibrated schedules of income and savings tax rates—which introduces a small amount of heterogeneity across incomes. The resulting across-income profile of s_{inc}' is plotted as the dashed red line in Figure 2.

Our second strategy for estimating s_{inc}' utilizes survey evidence about respondents' marginal propensity to save out of additional income. We draw on survey data from

[38] Our extension to many goods (online Appendix B.D) shows how the inclusion of cross-base responses affects optimal savings tax formulas. It could be used to compute the optimal savings tax on different savings vehicles, if there was a larger body of empirical evidence on savings elasticities and cross-base responses.

[39] Two other recent studies point to the promise of estimating such causal marginal propensities in a variety of settings. Golosov et al. (2021) study the response to lottery prize winnings in the United States, although the absence of third-party administrative reporting of wealth in the United States complicates the measurement of marginal propensities to save. Straub (2018) estimates the propensity to save out of permanent income, although the absence of quasi-experimental variation in earnings makes it difficult to separate causal income effects from across-income heterogeneity.

two sources. First is the Survey of Consumer Expectations, a monthly longitudinal survey of Americans conducted by the Federal Reserve Bank of New York. Since 2015, this survey has asked respondents to describe how they would allocate an unexpected 10 percent increase in their posttax income across spending, saving, or debt reduction. Because the Survey of Consumer Expectations also asks about income and assets, this provides a measure of s'_{inc} across incomes and savings. Second, to explore whether the resulting MPC is sensitive to the change in income being posttax or temporary, we also fielded a new probability-based survey representing the US adult population, conducted on the AmeriSpeak panel in the spring of 2021 (Lockwood et al. forthcoming; Ferey, Lockwood, and Taubinsky 2024). We asked respondents how much more they would save each year if they received a hypothetical raise that increased their household's annual income by $1,000 in the coming years. This question has the advantage of asking directly about a modest, *persistent* change in *pretax*, *earned* income, rather than a large one-time windfall, consistent with the definition of s'_{inc}. These two sources of survey evidence provide similar results, suggesting a short-run marginal propensity to save out of pretax income around 0.60, close to that reported in Fagereng, Holm, and Natvik (2021), with little variation across incomes. We translate these measures into long-run MPCs using the response profile of Fagereng, Holm, and Natvik (2021). The resulting profiles of $\overline{s'_{inc}}(z)$ are displayed as the solid and dot-dashed red lines in Figure 2.

These estimation strategies provide remarkably consistent estimates of $\overline{s'_{inc}}(z)$. We use the profile from our AmeriSpeak survey—the middle of the three similar profiles in Figure 2—as our baseline measure of $\overline{s'_{inc}}(z)$. There is a substantial difference between the across-income savings slope $\overline{s}'(z)$ and the causal income effect $\overline{s'_{inc}}(z)$, suggesting that across-income heterogeneity $\overline{s}'(z) - \overline{s'_{inc}}(z)$, plotted as a dotted line, rises with income and is positive across much of the income distribution.[40]

By way of comparison, Golosov et al. (2013) measure preference heterogeneity by estimating differences in discount factors across ability levels. They infer discount factors from a simple parametric model of savings choice applied to survey data on individuals' household income paths and net worth, and they use survey respondents' performance on the Armed Forces Qualification Test (AFQT) as a proxy for ability. In contrast to our findings, their estimation strategy finds very little measured preference heterogeneity, amounting to less than 1 percent of the cross-sectional variation in savings (see online Appendix E.A.3). This discrepancy could be driven by attenuation bias due to measurement error in their proxy for ability—an issue we avoid by computing preference heterogeneity as a difference of two statistics rather than from regression analysis. It could also be driven by their use of a narrower measure of across-income heterogeneity based only on time preferences, as opposed to all of the possible forms of heterogeneity that our statistic comprises.

[40] Our measure of across-income heterogeneity is slightly negative at low incomes, which in our simulations gives rise to savings subsidies at low incomes.

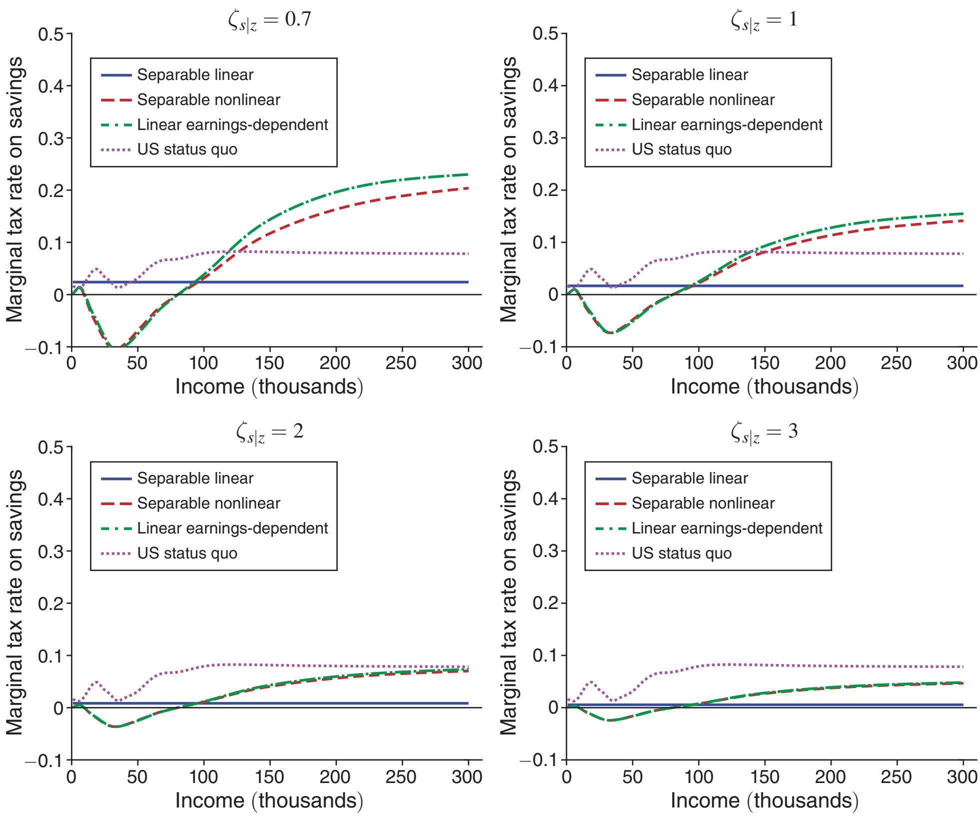

FIGURE 3. SAVINGS TAX RATES IMPLIED BY PARETO EFFICIENCY FORMULAS

Notes: This figure presents the marginal savings tax rates values that satisfy the Pareto efficiency formulas in Proposition 3, plotted against the earnings level to which they apply. We plot these schedules for four different values of the savings elasticity $\zeta_{s|z}$, with $\zeta_{s|z} = 1$ representing our baseline case.

C. *Results*

We first report results for the case of unidimensional heterogeneity. Figure 3 plots the schedule of marginal tax rates for SL, SN, and LED tax systems that satisfy the Pareto efficiency formulas in Proposition 3, taking the existing US income tax schedule and income distribution as given.[41] In each case, we translate the tax into a marginal tax rate on gross savings at retirement, measured in period 2 dollars. Each panel reports results for a different value of the savings elasticity. For SL tax systems, the linear savings tax rate τ_s is by definition constant across earnings levels. For LED tax systems, the linear savings tax rate $\tau_s(z)$ is earnings dependent, and we thus report the linear savings tax rate at each earnings level. For SN tax systems, the nonlinear savings tax schedule $T_s(s)$ depends on the value of savings s and not on earnings z. But to make the SN system visually comparable to the other systems,

[41] An alternative interpretation is that these schedules represent *optimal* (not just Pareto-efficient) savings tax rates under welfare weights that render the existing income tax optimal. We discuss these "inverse optimum" welfare weights in online Appendix E.B.2.

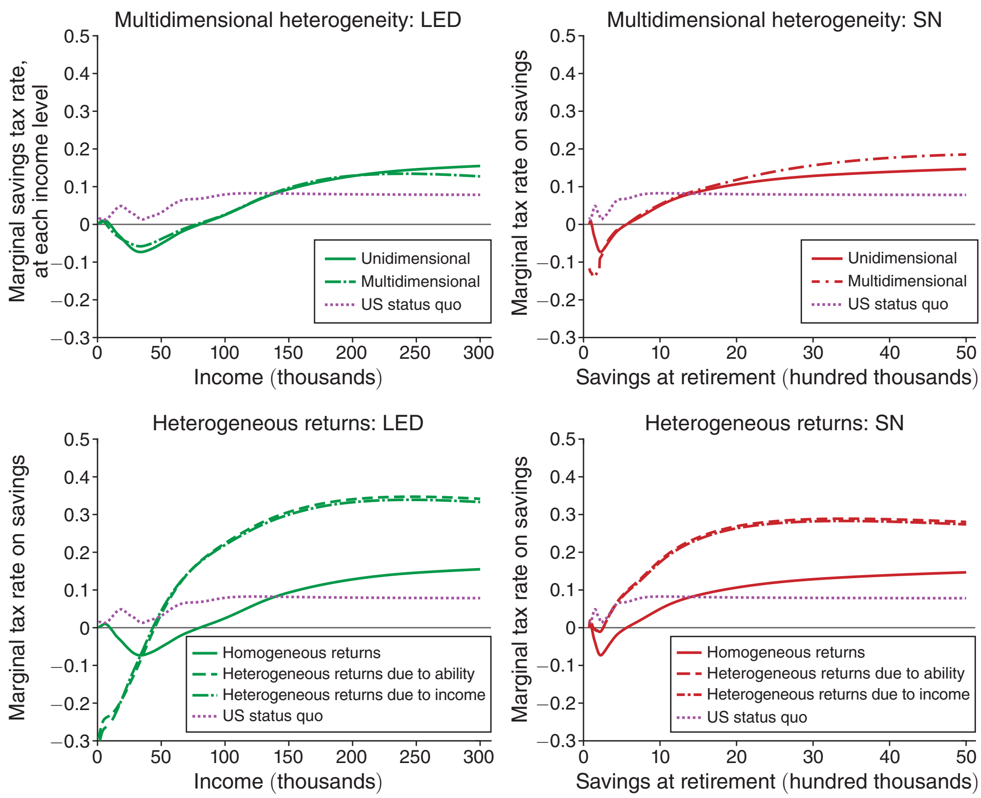

Figure 4. Effects of Multidimensional Heterogeneity and Heterogeneous Returns

Notes: This figure plots the marginal savings tax rate schedules that are optimal, according to the first-order condition formulas presented in the text, for two extensions discussed in Section VI: multidimensional heterogeneity (top row) and heterogeneous returns (bottom row). All plots also reproduce the Pareto-efficient savings schedules from Figure 3 for comparison, as well as the status quo US savings taxes. These plots use the same set of social welfare weights, calibrated to rationalize the status quo income tax in the unidimensional model. The Linear Earnings-Dependent (LED) schedules, in the left column, are plotted across earnings during work life. The Separable Nonlinear (SN) schedules, in the right column, cannot be plotted this way because individuals with a given income have different levels of savings and are thus subject to different savings taxes. We therefore plot them over total savings at the time of retirement. See Section VII and online Appendixes E.B and E.C for details.

we plot the marginal savings tax rate faced at the margin by each earner, given their level of saving represented in online Appendix Figure A1.

In each panel, marginal savings tax rates are positive for those earning above about \$80,000, and the nonlinear tax schedules are progressive, with marginal rates increasing with income across most of the distribution. This progressivity is shaped by the evolution of the sufficient statistic $\overline{s}'(z) - \overline{s'_{inc}}(z)$ along the income distribution and in particular by the contrast between the convex shape of the savings profile and the flatness of the causal income effect. The magnitudes depend on the value of the savings elasticity parameter. In the baseline case of $\zeta^c_{s|z} = 1$, savings tax rates in SN and LED tax systems are negative and small in magnitude for annual incomes below \$50,000, then they steadily increase to nearly 15 percent for annual incomes around \$200,000, remaining stable thereafter. Changing the savings elasticity parameter scales the savings tax rates without affecting the overall pattern:

across-income heterogeneity calls for (mostly) progressive savings tax rates, which are positive at higher earnings levels. At our baseline elasticity value, our estimates of optimal savings tax rates on high earners are around twice as high as the prevailing savings tax rates in the United States, which are also shown in Figure 3.

Figure 4 considers two key extensions to these results: multidimensional heterogeneity, as in Section VC, and heterogeneous rates of return with "tax arbitrage" efficiency effects, as in Section VIB. For comparability with our baseline results, these calibrations hold fixed other parameters from our baseline specification, including elasticity parameters and inverse-optimum welfare weights (see online Appendix E.B.2). These results are computed using our baseline savings elasticity of $\zeta^c_{s|z} = 1$. We plot both types of nonlinear tax schedules, LED and SN, omitting the separable linear plots for legibility.

In the case of multidimensional heterogeneity, we use the same measure of gross savings, but rather than compute average savings at each income, we partition the population into four levels of savings at each level of income, representing quartiles of the income-conditional savings distribution.

In the case of heterogeneous rates of return, we follow Gerritsen et al. (2020), who, relying on empirical work by Fagereng et al. (2020), assume that rates of return rise by 1.4 percentage points from the bottom to the top of the income distribution. We linearly interpolate this difference across income percentiles, centered on our 3.8 percent baseline rate of return.

The top two panels of Figure 4 show that incorporating multidimensional heterogeneity has only a small effect on optimal LED and SN tax rates. The reason for this can be understood from the formulas for optimal simple tax systems in Proposition 4: a key additional term involves the covariance between s'_{inc}—which scales the size of the earnings distortions from taxing s—and the fiscal externalities from earnings adjustments. In our setting there appears to be little heterogeneity in s'_{inc} (see online Appendix Figure A4 and the discussion in online Appendix E.B), and the remaining covariance terms turn out to be quantitatively small.

The bottom two panels show that heterogeneity in rates of return tends to significantly raise optimal savings tax rates, reflecting the efficiency effects of tax arbitrage highlighted in Proposition 6.[42] The bottom right panel shows that tax rates in the SN system are higher at all levels of income, consistent with our discussion of the formula for SN systems in Proposition 6. On the other hand, recall that the formula for LED systems implied lower savings tax rates at low incomes and higher saving tax rates at higher incomes. Consistent with this, the bottom left panel shows that relative to the baseline, the optimal savings tax rates with heterogeneous rates of return are even more progressive. For example, substantial savings subsidies are optimal for incomes below about $40,000, whereas savings taxes are substantially higher at higher incomes.

Taken together, our empirical results show a robust role for progressive savings taxes, stemming from across-income heterogeneity. This highlights the importance of this statistic and motivates additional empirical work estimating the long-run marginal propensity to save out of earned income, as well as across-income consumption profiles and causal income effects in other applications.

[42] Consistent with the tax arbitrage interpretation, these efficiency effects are (almost) unaffected by whether return heterogeneity is driven by income (scale dependence) or by ability (type dependence).

VIII. Conclusion

This paper characterizes optimal tax systems on earnings and savings (or other dimensions of consumption) in the presence of general across-income heterogeneity, both in unidimensional and multidimensional settings. We then derive formulas that characterize the optimal smooth tax system through familiar empirical statistics as well as a key sufficient statistic for across-income heterogeneity. This statistic captures heterogeneity in preferences, heterogeneous rates of return, and income-shifting abilities and spans some models of bequests and human capital acquisition. This generality highlights the value of empirical research estimating (long-run) marginal propensities to save and consume. We then use the same sufficient statistics to characterize a set of "simple" separable tax systems that are widely used in practice. We also provide tractable extensions to multiple goods, corrective motives, and heterogeneous prices with "tax arbitrage" efficiency effects. Finally, we apply our theoretical formulas to the setting of savings taxes in the United States. Results suggest that the savings tax rates that would be consistent with the existing income tax are (mostly) progressive and above prevailing rates at higher levels of earnings. Together, these results provide a practical and general method for quantifying optimal tax systems for savings, inheritances, and other commodities in the presence of general across-income heterogeneity.

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